Recovering the Designer's Ledger: Inverse Hedonic Pricing of Card Costs in a Collectible Card Game

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***Abstract*— Collectible card games (CCGs) price every card in a single integer resource, yet the rules mapping card text to cost are rarely published. We cast their recovery as an inverse hedonic pricing problem and propose ZHIZHI (Zero-residual Hedonic Inversion, Zero-shot Hedonic Imputation): Stage 1 inverts effect prices from anchor cards that differ in a single effect, so most prices are identified exactly; Stage 2 imputes any card's value from its text without refitting. The value-to-cost ratio is reported as the AMBER score (Attribute-Matched Budget-Efficiency Ratio), with 1.00 as par and repeatable triggers booked at half an activation. Applied to *Riftbound*, Riot Games' *League of Legends* card game, the framework fully prices 497 of 702 costed units and spells and 69 of 107 gear cards, extending a linear additive core with mechanism models for hand cards, selection and on-play effects. The table exposes latent design regularities: a 50% rule for repeatable contingent value (one-off conditions mostly at zero), free play-timing keywords, a 1 C same-target complementarity premium, and integer-rounding absorption of small effects on cheap units. The AMBER score has median 1.00 and interquartile range 0.86–1.07 (0.80–1.09 excluding the 146 anchor cards, at par by construction); calibrated prices predict held-out cards; and a nested ablation attributes nearly all error reduction to on-play adjustments. Booking triggers at half an activation improves prediction card by card (leave-one-out RMSE 1.541 to 1.400 C, factor fixed); its out-of-sample test is preregistered. We discuss transfer to other CCGs and use as a soft screen in procedural card generation.

***Index Terms*—** collectible card games, game balancing, inverse hedonic pricing, hedonic imputation, AMBER score, order statistics, inverse problems, robust statistics, reverse engineering, Riftbound

I. Introduction

Every card in a collectible card game (CCG) carries two things: a printed cost and a description of what the card does. The relation between them is decided by the designer, who works from an internal costing model—how much a point of damage is worth, how much a drawn card is worth, how much a keyword should cost—that is never published. Players infer fragments of this model from experience ("two mana for a 2/2 is fair"), and the game-AI literature has learned to *predict* printed costs from card text with black-box models [1, 2]. What neither approach delivers is the model itself: an explicit, interpretable price for each effect, with the evidence that supports it.

This paper recovers such a model for *Riftbound*, the *League of Legends* trading card game published by Riot Games [3], from static card data alone. We view the task as an inverse problem. The designer's costing function maps an effect bundle $\mathbf{x}$ to a cost c; we observe many $(\mathbf{x}_i, c_i)$ pairs, plus the structural constraint that costs are integers and combat statistics are integers, and we wish to identify the latent price vector. This is the hedonic pricing problem of Lancaster [4] and Rosen [5]—a good's price decomposed into implicit prices of its characteristics—applied to a market with a single, deliberate price-setter rather than competitive equilibrium. We solve it in two stages, which we call ZHIZHI (Zero-residual Hedonic Inversion, Zero-shot Hedonic Imputation).[1] Stage 1 inverts effect prices from anchor cards that differ in a single effect, so that most prices are identified exactly, with zero residual, rather than fitted; Stage 2 imputes the value of every card in the pool from its parsed text with the frozen price table, without refitting—the operation that price-index statistics call *hedonic imputation* [6], applied card by card.

We distinguish three layers of fairness in a CCG. *Rule fairness* is guaranteed by the rules themselves. *Value fairness* concerns whole decks interacting under hidden information and is, at present, computationally out of reach for a full card pool. *Unit fairness* asks whether a single card's statistics and effects are commensurate with its cost. Our target is the third layer only, and our measure of it is the AMBER score (Attribute-Matched Budget-Efficiency Ratio): the ratio of a card's imputed value to its printed cost, with repeatable triggers booked at the half activation the designer implicitly charges for (Section III-D) and 1.00 as par. Consequently, the object we recover is the designer's cost logic, not the card's competitive strength: a card can be priced exactly "at par" by the designer and still be strong or weak in play because of synergies, tempo, or metagame context.

Two uses motivate the work. First, archetypes differ widely in flexibility and ceiling, and the model identifies cards that received a surplus numeric budget from the designer—candidates for constructed play with unusual upside. Second, it expresses

[1] ZHIZHI and AMBER are also the names of the authors' two cats.

most of the information on the board in one unit, so that an in-game trade of process resources (might, damage, cards, runes) can be judged as favourable or not; Section VIII explains why this second use does not extend to points.

A. Contributions

1. **The ZHIZHI framework.** Simulation-free recovery of designer prices from static card text via zero-residual anchor pairs (Stage 1), followed by zero-shot imputation for every card in the pool (Stage 2). Each recovered price carries a confidence tier tied to the number of independent supporting cards.

2. **Micro-founded closed-form mechanisms.** Order-statistics pricing of selection (PREDICT, look-N-take-1, choose-one-of-X) [7, 8] and asymmetric mental accounting of hand stock (draw, discard, FLOW) [9, 10], each yielding predictions that can be checked on cards outside its calibration set.

3. **Empirical design regularities.** A 50% rule for uncertain value, zero price for timing keywords, a same-target complementarity premium, and integer-rounding absorption at low cost.

4. **The AMBER score and its validation.** An interpretable per-card balance metric with hold-out predictions, a nested ablation with leave-one-out cross-validation, and a parameter audit; over 497 fully priced units and spells the score has median 1.00 and interquartile range 0.86–1.07 (0.75–1.00 nominal; 0.80–1.09 among the 351 cards that are not anchors), and booking repeatable triggers at half an activation improves prediction card by card under leave-one-out re-estimation of the prices, with the factor itself fixed. The dataset, per-card valuations and anchor tables will be released as a separate dataset on Zenodo.

II. RELATED WORK

A. Game Balance and CCG Analytics

Game balancing is a broad and loosely defined design goal [11]. Computational treatments typically measure balance through play: restricted-play agents [12], automatic balancing by evolutionary search over game parameters [13], and—for CCGs specifically—evolutionary deck construction and playtesting in *Hearthstone* [14, 15], evolving card attributes to move matchup win rates toward parity [16], and Monte Carlo search for *Magic: The Gathering* (MTG) [17]. Hoover et al. [18] survey the AI challenges that CCGs pose. These works evaluate balance *ex post*, by simulating games; we instead recover the *ex ante* costing rules that the designer used to set printed values.

Closest to our aim is the "resource scale" work of Zuin and colleagues [1, 2], which trains text-representation networks and gradient-boosted trees to recommend the mana cost of an MTG card from its rules text, statistics, colour and rarity, and adds explainability tools that highlight cost-driving words. Bertram et al. [19] learn generalised card representations that predict human draft picks, a revealed-preference proxy for card strength. Both learn a mapping from card to cost or strength; neither yields an explicit price per effect with provenance. Summerville and Mateas [20] generate MTG cards with sequence-to-sequence models, and Ham [21] discusses rarity as a balance lever in collectible games. Our contribution is complementary: an interpretable price table that such generators could use as a constraint.

B. Hedonic Pricing and Mental Accounting

Lancaster [4] reframed consumer goods as bundles of characteristics; Rosen [5] formalised hedonic prices as implicit prices of those characteristics recovered from market data. We adopt this decomposition but replace the competitive market with a single designer, which changes the estimation problem in two ways: prices are set by rule rather than by equilibrium, so anchor pairs can identify them exactly; and costs are integer-valued, so rounding is a first-order source of residual variance. Thaler's mental accounting [9, 10] supplies the language for our treatment of hand cards, whose stock is priced asymmetrically while the act of playing is not. Price-index statistics use *hedonic imputation* for the reverse operation—pricing a product from its characteristics with previously estimated implicit prices [6]; Stage 2 of ZHIZHI is that operation applied card by card.

C. Selection Value and Order Statistics

The value of choosing the best of several uncertain alternatives is classical: Stigler [22] and Weitzman [23] for search, McFadden [24] for the expected maximum in discrete choice. For normally distributed candidates the expected maximum of n draws is tabulated by Harter [8] and treated generally by David and Nagaraja [7]; we use these constants to price "look at N, keep one" and modal effects with one calibrated dispersion parameter.

D. Robust and Small-Sample Statistics

Because most parameters rest on 3–15 cards, we rely on medians and the Hodges–Lehmann estimator [25], minimax (Chebyshev) fits alongside least squares in the spirit of quantile regression [26], the Mann–Whitney test [27], permutation tests, and hold-out prediction [28], following the general programme of robust statistics [29].

III. GAME BACKGROUND AND PROBLEM FORMULATION

A. Riftbound in Brief

Riftbound [3] is a two-resource CCG. Each card has an *energy* cost (the large number in the corner) and may additionally require *power*, shown as rune icons of one of six *domains* (Fury, Calm, Mind, Body, Chaos, Order). Runes are channelled from a separate deck and can be *exhausted* or *ready*. *Units* have a *might* value, fight at *battlefields*, and score points by *conquering* and *holding* them; *spells* resolve once; *gear* is a reusable permanent, some of it *equipment* that attaches to units; *legends* and battlefields have no cost. Timing is governed by keywords: ACTION

cards can be played on the owner's turn or during *showdowns* (combat); REACTION cards can be played "any time, even before spells and abilities resolve"; HIDDEN cards are placed face-down for one power and later played as reactions for zero; REPEAT lets a spell resolve twice for an additional cost. Other keywords used below include ACCELERATE (pay extra to enter ready), ASSAULT/SHIELD ($+X$ might while attacking/defending), TANK, BACKLINE, DEFLECT, GANKING (may move between battlefields), AMBUSH, TEMPORARY, DEATHKNELL (effect when the unit dies), FLOW (play from trash for a Flow cost, then banish), EMPOWER/EMPOWERED, VISION and PREDICT (look at the top card(s), optionally recycle), LEVEL/XP, HUNT, LEGION, BUFF, BURN, WEAPONMASTER, and QUICK-DRAW.

B. Data

The card pool was exported from the Rift Atlas database [30]: 923 cards with cost, might, domain, set, keywords and full rules text across five sets—*Origins* (OGN), *Spiritforged* (SFD), *Unleashed* (UNL), *Vendetta* (VEN), and the starter set *Proving Grounds* (OGS). The export was taken from the Simplified Chinese release, which preceded the English release; card names in this paper follow the official English printings, and collector numbers are given so every anchor can be checked. Rules text was parsed into effect annotations and priced per effect type; cards whose parse was incomplete were archived separately and never used as anchors. Table I(a) summarises coverage by card class; Table I(b) compares the fully priced, partially priced and deferred units and spells.

Gear, legends and battlefields. Gear is priced on the same C scale with one extra parameter, the number of uses the designer expects. An equipment card's play cost is compared with its attached value (the median difference over 28 fully priced equipment cards is 0), and reusable gear is valued at a design baseline of three uses if it enters ready, two if it enters exhausted or triggers on a condition, and one if TEMPORARY, with a judgemental factor of 1, 0.5 or 0.25 for narrower conditions. The 69 fully priced gear cards have median $\eta = 1.00$ and are ranked in the value table alongside units and spells; because the expected-use parameter is not tested by the ablation of Section VII-C, the statistics in this paper cover the 497 units and spells only. Legends and battlefields have no cost and hence no ratio; for them we report per-use net value (46 of 49 legends) and per-trigger yield (61 of 64 battlefields) as aids to in-game decisions, not for ranking.

Effects whose value depends on quantities outside the card are excluded from pricing: effects that scale with a unit's might (mutual damage, damage equal to might), counters and control theft (whose value depends on an unknown opposing card), and effects on all or an indefinite number of targets. Symmetric effects are booked at zero, and a card consisting only of symmetric effects is deferred. Ban status is not a pricing criterion (Section IV-E): three banned spells and two banned battlefields that had been set aside under an earlier version of the deferral rules remain outside the statistics, so that the preregistered frozen model is unchanged, while the other banned cards are priced like any other and flagged in the data (Section VI-I). Each deferred card is assigned to the first applicable rule; the counts by rule are in the notes to Table I. All units and spells were re-audited under the final rule set: 88 cards had a value or status revised (mainly 0–3 trigger bins for repeatable effects, full-trigger accounting of conditional effects, optional additional costs booked on the cost side, and reclassification of untargetable or all-target effects), and 28 previously unpriced spells were added; a per-card change log accompanies the data.

Scope of the priced sample. The statistics of this paper describe cards whose effects decompose into priced components. Deferral is mechanical—a card is deferred by the rules above before any value is computed—but it is not random: Table I(b) shows that fully priced units and spells are cheaper than partially priced and deferred ones (median C_{tot} 4 against 5 C; Kruskal–Wallis $p < 0.001$), more often common (43% against 11–13%) and single-domain, and less often spells (28% against 56% among deferred cards); the deferred group over-represents *Origins*, *Spiritforged* and two-domain cards. Conclusions such as the median AMBER score of 1.00 therefore apply to the priced population, in which expensive rare spells with pool-wide, might-dependent or counter effects are under-represented.

C. Problem Statement

Let card i have printed energy E_i, power P_i, might m_i (zero for spells), and an effect vector $\mathbf{x}_i \in \mathbb{R}^K_{\geq 0}$ counting K effect types. Let $\mathbf{v} \in \mathbb{R}^K$ be the designer's latent prices in a common unit. The designer's balance condition, up to integer rounding and deliberate deviations, is

$$m_i + \mathbf{v}^\top \mathbf{x}_i \;\approx\; C_{\text{tot}}(E_i, P_i), \tag{1}$$

where C_{tot} converts the two resources to the common unit. Given $\{(E_i, P_i, m_i, \mathbf{x}_i)\}_{i=1}^n$, identify C_{tot} and $\mathbf{v}$, and characterise the structure of the residuals. This is a parameter-identification inverse problem: the forward model (designer's costing) is unknown, the observations are its outputs, and identifiability comes from the combinatorial structure of the card pool (many cards share effect types in different combinations) rather than from randomised variation.

Two features distinguish it from a standard regression. First, identification is often *exact*: a pair of cards differing in a single attribute pins its price without error, so exact anchor pairs are preferred to global fits and global fits are used only where no such pair exists. Second, the response is integer-valued, and for units so is the might term; rounding therefore contributes a structured, one-sided residual of less than 1 C (under the floor convention of Section IV-C) that must be modelled rather than averaged away.

D. The AMBER Score

For a priced card the value $V = m + \mathbf{v}^\top \mathbf{x}$ is evaluated at the card's *base* scenario, conditions unmet and repeatable triggers fired zero times, giving V_0; for a unit with a repeatable trigger, V_1 is the value after exactly one trigger. The *nominal* value/cost

TABLE I
CARD POOL, VALUATION COVERAGE AND SCOPE OF THE PRICED SAMPLE

(a) Coverage by card class

Class	Type	n	Full	Partial	Deferred
Costed; η defined, ranked, ablated	Unit	469	359	51	59
	Spell	233	138	19	76[a]
	Subtotal	702	497 (70.8%)	70 (10.0%)	135 (19.2%)
Costed; expected-uses model, not ablated	Gear[b]	107	69	19	19
Zero cost; no η, not ranked	Legend	49	46 per-use net values		3
	Battlefield	64	61 per-trigger yields[c]		3[d]
Excluded	Reminder card	1	Buff-token reminder, not a playable card		

(b) Fully priced, partially priced and deferred units and spells

	Full $n = 497$	Partial $n = 70$	Deferred $n = 135$	p
C_{tot}, median (IQR)	4 (3–6)	5 (4–7)	5 (3–8)	< 0.001
Spells, %	27.8	27.1	56.3	< 0.001
Set, %				0.008
Origins	31.4	24.3	39.3	
Spiritforged	20.9	12.9	28.9	
Unleashed	25.4	31.4	16.3	
Vendetta	19.5	28.6	12.6	
Proving Grounds	2.8	2.9	3.0	
Domain, %				0.001
Fury	17.9	12.9	9.6	
Calm	15.1	17.1	15.6	
Mind	15.9	8.6	14.1	
Body	15.3	12.9	17.0	
Chaos	13.9	24.3	17.0	
Order	17.5	17.1	9.6	
Two domains	4.4	7.1	14.1	
None (tokens)	0.0	0.0	3.0	
Rarity, %				< 0.001
Common	42.7	11.4	13.3	
Uncommon	22.5	22.9	21.5	
Rare	20.7	41.4	37.0	
Epic[e]	14.1	24.3	28.1	

[a] Deferred units: effects on all or an indefinite number of targets (16), gear- or legend-dependent effects (12), effects scaling with might (12), play from the trash (6), zero-cost tokens (4), rule-changing effects (4), free play of other cards (2), TEMPORARY or copy effects (2), deferred by decision (*Jinx, Demolitionist* (OGN-030)) (1). Deferred spells: effects on all or an indefinite number of targets (21), effects scaling with might (16), counters (11), symmetric effects (7), control theft (5), effects with no current price (3), banned cards set aside (3; Section VI-I), gear- or legend-dependent effects (3), play from the trash (2), free play of other cards (2), rule-changing effects (1), untargetability (1), TEMPORARY or copy effects (1). Each card is deferred by the first applicable rule; the per-card list accompanies the data.
[b] 106 gear cards plus one unit gear (*Patched Porobot* (VEN-058)).
[c] 14 direct, 19 conditional, 25 symmetric (both players), 3 controller cost reductions.
[d] 2 banned (set aside; Section VI-I), 1 not computed.
(b): percentages are within column. p: Kruskal–Wallis for cost; for categories, χ^2 test of independence with a permutation p-value (20 000 draws; the asymptotic p-values agree to within 0.001). The four zero-cost tokens are excluded from the domain test. [e] Includes three alternate-art cards.

ratio is

$$\eta \;=\; \frac{V_0}{C_{\text{tot}}}, \tag{2}$$

and the AMBER score (Section I) books a repeatable trigger at a fixed fraction of an activation,

$$\eta^{\star} \;=\; \frac{V_0 + k\,(V_1 - V_0)}{C_{\text{tot}}}, \tag{3}$$

with $k = 0.5$ for a repeatable trigger, $k = 0.25$ for a trigger that carries an independent condition, and $k = 0$ otherwise. Unless stated otherwise, the AMBER score values repeatable triggers at the designer's implied rate of 0.5 activations (halved again when the trigger carries an independent condition); the nominal AMBER score η sets them to zero.[2] The factor 0.5 applies to units whose repeatable trigger has a positive single-trigger increment $V_1 - V_0$: 162 of the 497 fully priced cards, all of them units. It is halved to 0.25 for the 17 of them whose trigger sentence requires, beyond the triggering event, an independent state or outcome (a rune count, an excess-damage threshold, a number of units present, hand or deck content, something else achieved the same turn); a narrow triggering event, a standing prerequisite such as "if I am at a battlefield," a high-hit-rate look-N-take-1 effect, a condition on entering rather than on triggering, and conquer or hold triggers themselves are not conditions. Spells, gear and all other cards have $k = 0$, so that $\eta^{\star} = \eta$. The evidence for both factors, and its circularity, is set out in Section VI-A.

Conditional bonuses ("if ..., then ...": an extra effect or an upgrade that depends on a state outside the card, such as the opponent's board, the rune count or the trash) stay at zero in the base scenario and at full value in the full-trigger reference. This is a tested convention, not an omission. We ran on them the same two checks as on the trigger factors (Sections VI-A and VII-D): the fraction of the conditional value that the printed cost pays for, $f^{*} = (C_{\text{tot}} - V_B)/\Delta_{\text{cond}}$, where V_B is the value with triggers booked as in Eq. (3) and Δ_{cond} the value of the conditional part, has median 0.08 over the 46 fully priced cards with $\Delta_{\text{cond}} \geq 1$ C (95% interval 0.00–0.50; 20 cards at exactly 0), and booking the

[2] In v1 of this preprint the ratio was reported without a name; in v2 and in hypotheses H1–H8 of the preregistration (Data Availability), "AMBER score" denotes the nominal AMBER score η. Since v3 the AMBER score uses the baseline convention $\eta^{\star}$.

bonus at one half does not improve leave-one-out prediction card by card (ΔRMSE -0.11 C, 95% interval -0.51 to $+0.34$), while booking it in full is significantly worse. The base scenario therefore keeps conditional bonuses at zero; the asymmetry between the half activation booked for repeatable triggers and the zero booked for one-off conditions appears to be the designer's own rather than an artefact of the score (Section VI-A). The score still understates the minority of cards whose conditional bonus was charged at about one half. A *full-trigger* reference value (all conditions met, optional additional costs booked on the cost side, self cost reductions at their maximum, DEATHKNELL at full value, repeatable effects at 1–3 triggers; the optional ACCELERATE payment excluded) is reported alongside but not used for ranking. The *effective* ratio η_{eff}, which would weight conditions by their in-game realisation rate, is outside the model. Throughout, $\eta = 1.00$ is par—a unit of measurement, not an average: anchor cards return $\eta = 1.00$ by construction, and $\eta > 1.00$ denotes a nominal budget surplus, i.e. the designer booked more value than cost at the unit-fairness level, not that the card is strong in play. A card with a high nominal ratio but ordinary results in play is one whose conditions are realised far less often than the nominal scenario assumes—a high-variance card on paper, not an exception to the model. The ratio matters because the total resource available in a match is approximately bounded: playing cards with $\eta > 1$ converts a fixed budget into more value (130 C of value from 100 C of cost, say), although tempo, scoring and exchanges mean that a value surplus is not the same as winning.

Notation. C_{tot} is the printed cost in C (Section IV-A); V_0 and V_1 the base and one-trigger values; $\eta = V_0/C_{\text{tot}}$ the nominal ratio and $\eta^\star$ the AMBER score; $\eta_{(t)}$ the ratio with a repeatable trigger booked t times (the bins of Section IV-G); $\eta_B = V/B(c)$ the ratio against the might budget $B(c)$ of Section IV-C; k the trigger factor, λ the DEATHKNELL factor, n^* the implied number of activations, and f^* the implied realisation fraction of a one-off conditional bonus of value Δ_{cond} (Section VI-A). Differences below the model's resolution of 0.3 C (Section IV-E) are not read as evidence.

IV. METHOD

ZHIZHI proceeds in two stages. *Stage 1, zero-residual hedonic inversion* (Sections IV-A–IV-F), recovers the price table: the unit of account, the linear additive core with its anchor pairs, the might budget, the mental-accounting and order-statistics mechanisms, and the fit criteria. Most prices are identified exactly by anchor pairs, with zero residual; parameters supported by several cards are fitted by least squares or minimax (Section IV-E), so Stage 1 is zero-residual where the card pool allows and a small-sample fit elsewhere. *Stage 2, zero-shot hedonic imputation* (Sections IV-G–IV-H), evaluates every card in the pool from its parsed effect vector with the frozen price table and no refitting; a card all of whose effects are priced is fully priced, one with an unpriced effect is partial, and a card caught by a deferral rule is deferred (Table I).

A. Unit of Account and Resource Conversion

All values are expressed in cost units C, with 1 energy = 1 C. The exchange rate for power is identified by three independent card equations—*Hextech Ray* (OGN-009), *Falling Star* (OGN-029) and *Punch First* (SFD-097), each of which pays a mix of energy and power for effects whose prices are already fixed in C by energy-only cards—and all three solve exactly to 1 power = 2 C. The rate is corroborated by *Vengeance* (OGN-229) (4 energy, 2 power) and *Blast of Power* (OGS-012) (6 energy, 1 power), which have different compositions, the same total under the rate, and the same effect (kill a unit), and by the price of channelling an exhausted rune (Section V-A). Hence

$$C_{\text{tot}} = E + 2P. \tag{4}$$

B. Linear Additive Hedonic Core (Stage 1: Inversion)

The value of a card is modelled as the sum of its priced components,

$$V = m + \sum_{k=1}^{K} v_k x_k, \tag{5}$$

with modal choices, conditional effects and other non-additive components modelled separately (Sections IV-F–IV-G). Prices v_k are identified by *anchor-pair elimination*: two cards that differ only in effect k give v_k directly; cards differing in two effects give a two-equation system, and so on. For example *Incinerate* (OGS-003) (2 C, deal 2) and *Final Spark* (OGS-022) (8 C, deal 8) give 1 C per point of damage and a zero intercept. A price is assigned the confidence tier *high* when two or more mutually independent cards agree, *medium* when it rests on a single method or few samples, *medium–low* when a single anchor card supports it, and *low* when the value rests on an assumption rather than an anchor. Because anchor cards return $\eta = 1.00$ by construction, agreement among anchors is not itself evidence of balance.

C. Piecewise-Linear Might Budget

Vanilla units (no effect text) satisfy might = cost up to 5 C, after which might grows faster than cost: *Mega-Mech* (OGN-088) is 7 C for 8 might and *Mountain Drake* (OGN-142) is 9 C for 10 might. Every vanilla unit satisfies might $= \lfloor B(c) \rfloor$ for the budget function

$$B(c) = \begin{cases} 1.1\,c, & c \le 5, \\ 5.5 + 1.33\,(c-5), & c > 5, \end{cases} \tag{6}$$

measured in C, since one might is defined to be worth 1 C. Under the floor convention the two slopes are identified only up to intervals: every slope in $[1.00, 1.20)$ reproduces the five vanilla units at 1–5 C, and every slope in $[1.25, 1.375)$ reproduces Mega-Mech and Mountain Drake; 1.1 and 1.33 are the values used in the value table. The rounded budget $\lfloor B(c) \rfloor$ is the same for every slope in these intervals at 1–9 C, so the might budgets, the DEATHKNELL fit and the separation of effect value below do not depend on the choice. The unrounded slack $B(c) - c$ does (at 4 C it ranges from 0 to 0.8 C over the first interval), and the three places that use the unrounded value—the on-play

multiplier of Section IV-H, the keyword-rounding boundary of Section VI-E and the might-only sensitivity check of Section IV-H—are marked as such. $B(c)$ is the total value the designer expects a c-cost unit to deliver. It is used (i) as a reference ratio $\eta_B = V/B(c)$ and (ii) to separate effect value on expensive units. The η denominator remains C_{tot}.

D. Mental Accounting of Hand Cards

A recurring difficulty in CCG valuation is whether a played card "costs a card." In the designer's ledger it does not. A per-card hand charge would appear in anchor pairs as a common non-zero intercept, and the damage pair *Incinerate* (OGS-003) / *Final Spark* (OGS-022) shows none: 1 C per point of damage with intercept 0 (Section IV-B); nor does the draw pair *Consult the Past* (OGN-083) (4 C, draw 2) / *Progress Day* (OGN-114) (8 C, draw 4): 2 C per card with intercept 0. Two independent effect types priced through the origin are the evidence for the rent-free reading, and the second pair is listed as a prediction of the first in Table IX. We read this as a mental-accounting model [9] that treats the per-turn draw as an exogenous base endowment: the *act* of playing converts a hand slot into board value rent-free, so playing costs exactly $C_{\text{tot}} = E + 2P$, and only effects that change the *stock* of hand cards are priced, asymmetrically, at the margin—creating a hand card is resource generation, draw $1 = +2.00$ C; losing one is resource loss, discard (the lowest-quality card in hand) $= -1.00$ C. The 1.00 C gap is the reserve price of a hand card in the designer's ledger. Under the adopted dispersion, the order-statistics model of Section IV-F implies 0.83–1.04 C for the lowest card in hand, consistent with the discard price (Section VII).

FLOW is interpreted in this framework: a Flow replay does not refund any hand rent; it grants one "discard avoided," worth $h = 1.00$ C, because the card would otherwise have stayed in the trash. This closes the accounting of Flow cards (Section V-D).

E. Estimation and Inference

1) Fit criteria: For parameters supported by several cards we compute both the least-squares solution and the minimax (smallest maximum absolute deviation) solution and choose a value between them. Ratio-type parameters k (e.g. the DEATHKNELL coefficient, target-restriction discounts) use regression through the origin,

$$\hat{k} = \frac{\sum_i X_i V_i}{\sum_i V_i^2}, \tag{7}$$

where X_i is the implied value recovered from card i and V_i the unconditional value of the same effect.

2) Robust statistics: Medians are the primary location statistic; the Hodges–Lehmann estimator [25] is reported where useful. Cards that deviate markedly from their peers are excluded as anchors and used only as bounds, with conclusions compared with and without them. Because bans usually reflect combinations rather than single-card value, ban status is not evidence in single-card pricing. No deck-composition proxies (e.g. pool-wide frequencies) are used.

3) Group comparison: To test whether an attribute is priced, cards are grouped by the attribute and the mean, standard error and 95% confidence interval of (cost − priced effect value) are computed; an interval excluding a hypothesised value rejects it.

4) Hold-out prediction: Parameters fitted on one group of cards are used to predict a disjoint group (e.g. the zero intercept of the damage pair predicts that draw cards are priced through the origin; the distribution of EMPOWER costs predicts the value of a free Empower). Results are in Section VII.

5) Resolution: Throughout we treat 0.3 C as the resolution of the model. It is a working threshold, not an estimated quantity: about a third of the 1 C cost step, and the size of the disagreements that the data leave unresolved between estimators of the same parameter (least squares against minimax: PREDICT 0.385 against 0.40 C per card, λ 0.46 against 0.54, gear removal 2.75 against 2.875 C) and of the typical half-width of the exact bootstrap intervals of multi-anchor parameters (Section VII-E). Differences below it are reported but not read as evidence for or against a rule.

F. Order-Statistics Model of Selection Premia

PREDICT, look-N-take-1 and "choose one of X" effects all let the player keep the best of several uncertain options. Model option i as $\mu_i + \sigma Z_i$ with standardised noise Z_i. The premium from choosing is

$$\Delta \;=\; \mathbb{E}\left[\max_i(\mu_i + \sigma Z_i)\right] - \max_i \mu_i. \tag{8}$$

For equal means and normal noise, $\Delta \approx 0.564\sigma$ ($n = 2$), 0.846σ ($n = 3$), 1.029σ ($n = 4$) [7, 8]. The dispersion σ is calibrated once per mechanism from cards whose other components are independently priced. The same model gives the expected value of the *worst* of n cards, which is what discard and "take one, trash the rest" effects remove.

G. Discrete Scenario Binning (Stage 2: Imputation)

For repeatable triggers ("whenever…," "when I move," activated abilities) and for count-dependent effects, we do not impose an external probability or deck model. Instead value and η are reported in bins—0, 1, 2 and 3 triggers—with the base scenario at zero. The ranking convention is the 0.5-activation baseline of Section III-D; the bins are still reported. Conditional effects are reported at two bounds, *base* (unmet) and *full-trigger*. Flow cards are reported under four regimes (Section V-D).

H. Separating Effect Value from Might

For spells, effect value is C_{tot} minus all other priced components. For units above 5 C, effect value is $B(c) - m -$ (other priced components), all in C; the implicit assumption is that the high-cost scale discount applies to the whole value budget rather than to might alone. This is supported by data: 19 units carrying a single on-play effect, separated in this way, yield effect values equal to $1.10\times$ (resource effects) and $1.12\times$

(removal effects) the price of the same effect on a spell, i.e. approximately full price. The excess over 1.00 is the fractional budget slack: under whole-card rounding $V = \lfloor B(c) \rfloor$, the six such units whose effect is integer-valued (*Stormclaw Ursine* (OGN-137), *Riptide Rex* (OGN-092), *Harnessed Dragon* (OGN-234), *Beast Below* (SFD-132), *Cemetery Attendant* (OGN-165), *Ocean Drake* (VEN-115)) match exactly, so the multiplier is kept at 1.00. Under the alternative (discount applies to might only, so surrendered might is converted at the marginal rate $1/1.33 \approx 0.75$ C per might), the same effects would be worth only $\approx 0.8\times$ spell price. For units at or below 5 C the sub-1 C budget slack is rounding and is not credited as effect value; effect value is simply $c - m$.

Sensitivity. The DEATHKNELL coefficient λ (the fraction of a Deathknell effect's unconditional value that is booked; Section V-F) is 0.50 (median) and 0.46 (least squares) under the whole-budget convention. Under the might-only convention (which converts the unrounded shortfall $B(c) - m$ at 0.75 C per might, whereas the adopted fit uses $\lfloor B(c) \rfloor$), *Ruined Rex* (UNL-067) moves from 0.75 to 0.66 and *Tasty Faefolk* (OGN-075) from 0.33 to 0.27, leaving the five-card median at 0.50 and least squares at 0.41; the conclusion $\lambda \approx 0.5$ is insensitive to the convention.

V. RESULTS: THE RECOVERED PRICE TABLE

A. Base Prices

Table II lists the core parameters, grouped by confidence tier, with the anchors that identify them. Several entries deserve comment. The price of a point of damage does not depend on whether the target must be at a battlefield. ASSAULT X and SHIELD X are worth 0.50 C per point because they apply only when attacking or defending, and the same rate applies at $X = 1$ and when granted to other units. Unconditional kill (8.00 C) is identified by two cards with different resource compositions and the same total, and it serves as the overflow cap for single-target damage and "$-X$ might this turn": across the pool only 9 units (1.9%) have might ≥ 9 and 6 have might ≥ 10, so *Moonlight Affliction* (UNL-066) (-10 might) is booked at 8.00. Returning an enemy unit to hand is 6.00 (*Rebuke* (OGN-172)); returning one's own is -1.00, cross-checked by *Retreat* (OGN-104) and *Star-Crossed* (UNL-128). Scoring one point is 11.50 C from six cards whose mean equals their median; Section VIII discusses why this value is a design-side threshold rather than an in-game exchange rate.

B. Predict and Look-N-Take-1

PREDICT N does not change hand size; it improves the quality of subsequent draws. The order-statistics model prices it with random-card quality fluctuating around the draw value of 2 C, with dispersion calibrated from the cost gap between *Clairvoyance* (VEN-056) (7 C, Predict 5 then draw 2) and *Consult the Past* (OGN-083) (4 C, draw 2). Jointly fitting Clairvoyance, *Apprentice Mage* (VEN-047) and the discard price gives 0.385 C per card (least squares) and 0.40 C (minimax); we adopt 0.40 C, and uniform versus normal noise changes the result by at most 0.05 C. Table III gives the derived values. VISION is Predict 1. A unit carrying Vision is credited the full 0.40 C at 4 C and above, and 0.12 C on a 2–3 C unit under keyword rounding (Section VI-E).

Look-3-take-1 can be read as "draw 3, then discard the worst two of the three new cards": under the same dispersion the three order statistics of three fresh draws sum to 6.00 C, so the reading is an identity rather than a check (the expected values of the two discards, 1.17 and 2.00 C, leave exactly the 2.83 C of Table III). The discard price of -1.00 C is the minimum of a whole hand, not of three fresh draws; the two follow from the same model and are not in tension.

C. Modal "Choose One" Effects

A modal card is worth its best single mode plus a selection premium. Calibrating $\sigma \approx 1.4$ C on *Qiyana, Victorious* (OGN-155) ($+0.85$ C), *Buhru Captain* (SFD-091) ($+0.50$ C) and *Disposal Order* (UNL-103) (0) gives, for equally valued modes, $+0.8$ C (two modes), $+1.2$ C (three) and $\approx +1.45$ C (four): the increment falls from ≈ 0.4 C to ≈ 0.25 C as modes are added. The premium shrinks with the gap between modes: $+0.4$ C at a 1 C gap and $+0.16$ C at 2 C. When the *opponent* chooses, the sign reverses: *Shakedown* (OGN-033) (modes worth 4 C and 6 C) is worth $4 - 0.16 = 3.84$ C against a printed 4 C.

D. Flow

A Flow card is played twice. The first play, from hand, pays the printed cost and consumes a hand card; the value k of the Flow keyword is booked here. The Flow play pays the Flow cost F and consumes no hand card, yielding the effect value V plus the saved hand card ($h = 1.00$ C, Section IV-D). On the printed side,

$$k = 0.2 \max\big(0,\, 1 - p\big), \qquad p = F - C_{\text{tot}}. \tag{9}$$

The coefficient is estimated from the five spells with $p = 0$, where implied $k = C_{\text{tot}} - V$: *Dredge Up* (VEN-049) (2 C, draw 1) gives 0; *Up from the Deep* (VEN-100) (3 C, two 1-might Tentacle tokens at 2.8 C) 0.2; *Iterative Design* (VEN-051) (4 C, a 3-might Mech token at 3.4 C) 0.6; *Onslaught* (VEN-081) (4 C, $+6$ might at the non-showdown discount, 3 C) 1.0; *Perfect Execution* (VEN-012) (5 C, effect 4.5 C including the same-target complementarity premium of Section V-J) 0.5. The median of the five is 0.5 C, the mean 0.46 C. However Onslaught's effect value depends on the non-showdown discount, which is itself supported mainly by Onslaught; using it to raise k would be circular, and without it the median of the remaining four is 0.35 C (mean 0.33 C).

The parameter cannot be pinned down precisely. The seven scored Flow cards with $p \geq 1$ (an eighth, *Twilight Shroud* (VEN-031), is unscored because it grants "can't be chosen") should show zero Flow premium in theory but range from -1.8 to $+2.5$ C (s.d. ≈ 1.4 C), which is the noise of the effect prices themselves; with that noise the four-card standard error is ≈ 0.7 C, and ± 0.25 C precision would need about 31 cards with $p = 0$. The $p = 0$ and $p \geq 1$ groups are statistically indistinguishable (exact Mann–Whitney $p = 0.52$). Independent evidence comes from REPEAT, which is also "repeat the effect without spending a hand card": of seven Repeat spells whose effect is priced by other

TABLE II

BASE PRICES RECOVERED BY ZHIZHI (STAGE 1), IN C, GROUPED BY CONFIDENCE TIER. H = TWO OR MORE INDEPENDENT CARDS AGREE; M = SINGLE METHOD OR FEW SAMPLES; M–L = SINGLE ANCHOR CARD; L = ASSUMPTION-BASED.

Effect	Price	Conf.	Identifying evidence
1 power (rune icon)	2.00	H	Hextech Ray, Falling Star, Punch First (exact); closes with rune channelling below
1 might	1.00	H	5 vanilla units; 18 further units with only zero-priced keywords
1 damage	1.00	H	Incinerate vs. Final Spark; no premium for battlefield-only targets
$+1$ / -1 might this turn	1.00 / 1.00	H	Guttural Roar (VEN-072), Punch First; Frigid Touch (SFD-066), Smoke Screen (OGN-093); both realised only through combat, $\times 0.50$ when usable only in the main phase (Table V)
SHIELD X, ASSAULT X	0.50 per point	H	Half price (conditional on attacking/defending); same at $X = 1$ and when granted; 0.15 when native to a 2–3 C unit (keyword rounding, Section VI-E)
Draw 1	2.00	H	Progress Day (OGN-114): 8 C for draw 4; Dredge Up, Downstage Dramatics, Consult the Past
Channel 1 exhausted rune	2.00	H	Mobilize (OGN-134), Catalyst of Aeons (OGN-138); Qiyana, Victorious "draw 1 or channel 1"
Gain 1 power	2.00	H	Same as paying it (no discount for restricted use or timing); Seal of Rage (OGN-040) and five other Seals, Honeyfruit (UNL-049)
1 Gold gear token	0.875	H	Trove Golem (SFD-174) 0.75 and Deadly Flourish (UNL-073) 1.00, midpoint
Discard 1 (own hand)	-1.00	H	Lunar Boon (UNL-125), Chemtech Enforcer (OGN-003); lowest-quality card in hand
ACTION, REACTION, REPEAT, HIDDEN, ACCELERATE, DEFLECT, WEAPONMASTER, AMBUSH, TANK	0	H	Multiple cards each; timing analysis in Section VI-B. The keywords themselves are free; a paid Accelerate prices "enters ready" at 2.00, the same as readying a unit
GANKING	1.00	H	Kinkou Lifeblade (VEN-093), Brutal Hunter (VEN-070) gain Ganking on Empower at exactly 1.00; innate mean $\approx$1.1
Move 1 enemy unit (any destination)	3.00	H	Charm (OGN-043); Skyward Strike (UNL-038), Void Assault (UNL-202), Moonfall (UNL-198)
Move 1 friendly unit	1.00	H	Flash (OGS-011), Void Assault (UNL-202), Shuriken Flip (VEN-140)
Ready a unit	2.00	H	Upstage Comedy (UNL-009), Wallop (OGN-146); Meditation (OGN-048) in reverse
Kill a unit (unrestricted)	8.00	H	Vengeance (OGN-229), Blast of Power (OGS-012); Hidden Blade (OGN-213) consistent
Return enemy / own unit to hand	6.00 / -1.00	H	Rebuke (OGN-172); Retreat (OGN-104), Star-Crossed (UNL-128)
Score 1 point	11.50	H	6 cards, mean = median
May be played to an open (or attacking) battlefield	$+1.00$	H	Sneaky Deckhand (OGN-176), Rengar, Pouncing (SFD-025), Ocean Drake (VEN-115) at 1.00; Sai Scout (OGN-174) 0.60
Remove a buff (additional cost)	-1.00	H	Symmetric with buff = 1.00
Survive: remove damage, exhaust and recall a unit	2.00	M	Zhonya's Hourglass (OGN-077, anchor), Tactical Retreat (UNL-175); Guardian Angel (SFD-051) consistent
Kill a gear (unrestricted)	2.75	M	Median of 4 unconditional cards (Section VI-G)
Gain 1 XP / spend 1 XP	0.40 / 0.50	M	18 XP-gaining units; XP-to-buff conversion (Section VI-D)
Ready a rune / gain N energy	1.00 / N	M	One rune = one energy
BACKLINE	0	M	As TANK, DEFLECT
Stun a unit	4.00 any / 2.00 restricted	M–L / H	Unrestricted: Rune Prison (OGN-050) alone (Table XIV); restricted: Thwonk! (SFD-040), Heroic Charge (UNL-155)
Opponent discards 1 (their choice)	1.00	M–L	Bewitching Spirit (UNL-121)
Return a unit from trash to hand	2.00	M–L	Morbid Return (OGN-170)
May be played to an occupied enemy battlefield	$+2.00$	M–L	Dauntless Vanguard (SFD-093); Deadbloom Predator (OGN-161) lies beyond the budget curve and is not used
Kill a friendly unit (as additional cost)	-2.00	M–L	Cruel Patron (OGN-208); consistent with Stalking Wolf (UNL-166), Commander Ledros (OGN-231), Atakhan (UNL-170)
Return or kill own gear (additional cost)	-1.00	M–L	Legion Quartermaster (SFD-044)
Opponent kills one of their own units (their choice)	1.40	L	Opponent picks the weakest; booked as one 1-might token
Can't be chosen by enemy spells/abilities	unpriced	—	High practical ceiling; no reliable anchor; cards carrying it are marked partial

cards (Downstage Dramatics, UNL-061; Upstage Comedy, UNL-009; Feral Strength, SFD-034; Frigid Touch, SFD-066; Bonds of Strength, SFD-151; Blood Rush, SFD-003; Desert's Call, SFD-031), six carry a Repeat premium of exactly 0 and one of -0.4 C. Since Flow can be deferred to later turns it should be worth at least as much as Repeat. The plausible range is thus 0–0.6 C;

TABLE III
SELECTION EFFECTS PRICED BY THE ORDER-STATISTICS MODEL

Effect	Value (C)
PREDICT N (VISION = Predict 1)	$0.40N$
Look 2, take 1 / look 3, take 1	2.55 / 2.83
Look 4, take 1 / look 5, take 1	≈3.00 / 3.12

we adopt the midpoint of the direct estimate (0.35 C without Onslaught) and the Repeat evidence (0), rounded to 0.2 C, and list it among low-confidence parameters. The parameter review of Section VII-E quantifies the uncertainty: on the four direct cards the exact bootstrap 90% interval of the midpoint rule is $[0.00, 0.28]$ (unshrunk median $[0.00, 0.55]$); fixing k at any value from 0 to 0.6 moves the full model's leave-one-out RMSE only in the fourth decimal; eight estimation rules, $k \equiv 0$ among them, have indistinguishable leave-one-out error on the four cards; and a joint fit of k and the truncation point over the eleven scored Flow cards other than Onslaught cannot identify the truncation. *Shuriken Flip* (VEN-140) with $p \geq 1$ recovers 0, consistent with the truncation in (9).

Each Flow card is reported under four regimes: printed $\eta = (V+k)/C_{\text{tot}}$; first play only V/C_{tot}; Flow play $(V+1)/F$; both plays $(2V+1)/(C_{\text{tot}}+F)$. When $F = C_{\text{tot}}$ the Flow play is the more resource-efficient (Dredge Up: 1.00 first, 1.50 on Flow); at $F = C_{\text{tot}}+1$ the two are roughly equal; at $F \geq C_{\text{tot}}+2$ the Flow play is less efficient.

E. Unit Tokens

A token unit is worth might +0.40 C: 13 usable samples (spells, on-play effects and Deathknell) have median +0.40 C, unchanged when Deathknell samples are removed. Within-type dispersion (Recruit tokens +0.25 to +1.55 C) exceeds between-type differences, so one formula is used. Mechanically special tokens are priced individually: the Shadow Clone token ≈3.9 C; a 3-might TEMPORARY Sprite entering ready ≈2.75 C.

F. Deathknell

A DEATHKNELL effect is booked at $\lambda \approx 0.50$ of its unconditional value. If it were booked in full, Deathknell units would systematically exceed their cost; the fit therefore solves for λ such that might $+\lambda\times$(unconditional value) equals the budget might $\lfloor B(c) \rfloor$, i.e. $\lambda = (\lfloor B(c) \rfloor - m)/V_{\text{DK}}$, where V_{DK} is the unconditional value of the Deathknell effect. For the single-Deathknell units: *Watchful Sentry* (OGN-096) $(2-1)/2 = 0.50$; *Soaring Scout* (OGN-216) 0.50; *Black Rose Dignitary* (UNL-152) $(3-2-0.15)/2 = 0.43$ (Assault 1 at 0.15 under keyword rounding, Section VI-E); *Ruined Rex* (UNL-067) $(9-6)/4 = 0.75$; *Tasty Faefolk* (OGN-075) $(8-6)/6 = 0.33$; *Undercover Agent* (OGN-178) $(8-5)/2 = 1.50$. Excluding the outlier Undercover Agent (a first-wave card), the five remaining samples give: least squares through the origin 0.46, minimax 0.54, median 0.50, mean 0.50 (t-based 95% CI 0.31–0.69). All four statistics fall in 0.46–0.54; we adopt $\lambda = 0.50$. *Honest Broker* (SFD-155) serves as a check: 2 C for 2 might plus a Gold token (0.875 C); at $\lambda = 0.5$ the value is $2 + 0.44 \approx 2.44$ C, which rounds to the printed 2 C. Tokens created by Deathknell are discounted at the same rate.

TABLE IV
DISCOUNT FACTORS FOR TARGET RESTRICTIONS (ZHIZHI STAGE 1)

Restriction	Factor	Evidence
At a battlefield only	1.00	Incinerate, Blast of Power
State/position (attacker, same place, pre-damaged)	0.50	Thwonk!, Heroic Charge, Facebreaker (OGN-220)
Destination restricted	0.67	Isolate (UNL-124), Temptation (SFD-129)
Kill, might ≤ 3	0.55	Soul Harvest, Lacerate (VEN-127), Sandshifter (SFD-158)
Return to hand, might ≤ 3	0.20	Gust (OGN-169), Windsinger (SFD-138)
Kill, might ≤ 2	0.15	Blood Money (single anchor; net of its Gold token)
Trait or relative-might restriction	0.50	Decree of Unity (VEN-131), Public Execution (VEN-154)

G. Target-Restriction Discounts

Restricted versions of an effect are priced as a fraction of the unrestricted price (Table IV). Restricted kill is priced *above* equivalent damage: *Soul Harvest* (UNL-159) (4 C, kill a unit with might ≤ 3 at a battlefield) is printed at 4 C, whereas 3 damage costs 3 C, even though 3 damage dominates in use (it also kills $\leq$ 3-might units and combines with combat damage). The model records the designer's price, so the nominal 4 C is retained; in expectation such cards should underperform damage spells of equal cost, an instance of nominal η exceeding effective η. *Blood Money* (SFD-162) (2 C: kill a unit with might ≤ 2 at a battlefield, plus a Gold token) shows no such premium: net of the token the kill is worth about 1.1 C, roughly the damage needed.

H. On-Play Effects: Cost-Controlled Regression

Units with "When you play me" (on-play) effects require two adjustments. First, effects that must be realised through combat in the same turn—$+X$ might this turn, stun, readying *other* units, granting a this-turn keyword—are worth less on a unit than on a spell: a spell with ACTION or REACTION can be played inside a showdown and realise its effect immediately, whereas a unit can be played only on its owner's turn outside showdowns, before combat, when it is unknown whether combat will occur; and a unit enters exhausted, so a this-turn bonus on itself is nearly unusable. Damage is *not* discounted: its value is settled at resolution (kill or waste) exactly as on a spell, and that risk is already priced into the spell anchors. "$-X$ might this turn" is discounted: it cannot kill by itself (a 0-might unit still needs 1 damage), so, like $+X$ might, it is realised only through combat

and takes the same factor. Second, cheap units price on-play effects at a small fraction regardless of type (Section VI-E).

The two effects are confounded: seven of the nine initial combat-trick anchors cost ≤ 4 C. We therefore regress, over 25 units with a single on-play effect priced independently by other cards, the recovered fraction of spell price on two indicators, low cost (2–3 C) and combat-trick type. The low-cost effect is -0.63 ($p < 0.001$) and the combat-trick effect -0.27 ($p = 0.057$); the best multiplicative fit is $0.25\times$ (low cost) and $0.55\times$ (combat trick). Optional-additional-cost anchors, in which the designer prices the effect directly and which are immune to might rounding, give 0.50 for *Masa, Crashing Thunder* (VEN-120), *Nami, Headstrong* (UNL-052) and *Kennen, Keeper of Balance* (VEN-135), with *Gust Monk* (VEN-101) at 1.00 as a single exception; because an optional mode is usually left with some surplus for the player, these values are lower bounds. Among expensive units *Whiteflame Protector* (OGN-082) gives 0.75. Taken together the combat-trick factor lies in 0.50–0.75, and the regression rejects 1.00 for the class as a whole; we adopt 0.50, consistent with the 50% rule of Section VI-A. (Sample sizes differ across the checks in this paper—19, 25, and $7+5$ units—because each uses the units that satisfy its own inclusion rule: single effect, effect priced independently elsewhere, effect type, and cost band.) Units with HIDDEN can be played as reactions but only where the player already has a unit, so they take the same factor (*Teemo, Scout* (OGN-197) recovers 0.33). *Frostcoat Cub* (SFD-067) (optional cost) and *Blastcone Fae* (OGN-097) (HIDDEN, on-play -2) are printed at the full $-X$ price, i.e. the designer then priced $-X$ might like damage; under the model's rule both come out conservative (full-trigger η 0.80 and 0.75). Neither is in the regression. Units that enter ready (priced by the ACCELERATE anchor), permanent effects (buffs, kills, moves) and already half-priced SHIELD/ASSAULT grants are not discounted again.

After both adjustments, fully priced units with $\eta \geq 1.5$ fell from 8 to 1 (the survivor, *Ol' Poro* (VEN-029), was later moved to partial status because "cannot be played on your first three turns" is a unique mechanic) and units with $\eta > 1.17$ from 29 to 13 (17 before keyword rounding, Section VI-E), while the median and quartiles (1.00; 0.75–1.00) were unchanged.

I. Other Mechanisms

Table V collects the remaining priced mechanisms. Repeatable effects ("whenever," "when I move," activated abilities, auras) are reported in 0–3 bins with the single-trigger value taken from the price table; an activated ability's per-use net value is its effect value minus its activation cost (energy 1 C, power 2 C, and -1 C if the cost includes exhausting the unit itself).

J. Same-Target Complementarity Premium

The additive core prices a spell as the sum of its parts. One class of spells departs from this in a consistent direction. Among the eight fully priced spells in which the price of "move a friendly unit" is the only unknown, the implied price of the move splits into two clusters: four spells whose other effect targets a different unit, or that have no other effect, all give exactly 1.00 C (*Flash* (OGS-011), *Shuriken Flip* (VEN-140), *Void Assault* (UNL-202), *Tactical Retreat* (UNL-175)); three spells that first reposition a friendly unit and then boost the *same* unit all give exactly 2.00 C (*Ride the Wind* (OGN-173), move then ready; *Showstopper* (OGN-270), buff then move; *Resonating Strike* (VEN-034), move then $+2$ might). No single value of the move price fits both clusters, whereas the structural split fits seven of the eight cards exactly (the remaining card, *Zenith Blade* (OGN-262)—7 C to stun an enemy and optionally move a friendly unit to it—implies 3.00 C for the move, 2 C above the different-target cluster; it is the card behind the "≥ 6 C" bound on the unrestricted stun in Table XIV). The same pattern appears in *Vault Breaker* (UNL-010) (grant ASSAULT 2 and GANKING to one unit: 2.00 C of parts for 3 C) and *Perfect Execution* (VEN-012) (ready a unit and grant it ASSAULT 3: 3.5 C of parts plus the Flow keyword for 5 C). All five cards carry a premium of 1.0–1.3 C over their additive value, and no spell of this shape is priced additively; spells whose positioning and boost act on different units (Shuriken Flip, Void Assault, *Heroic Charge* (UNL-155), *Moonfall* (UNL-198)) have zero residual.

We therefore book a *same-target complementarity premium* of $+1.00$ C for a spell that both positions a friendly unit (move, ready, or grant GANKING) and enhances the same unit for combat ($\pm X$ might this turn, a buff, ASSAULT/SHIELD). Four of the five carriers sit at exactly $+1.00$ and one (Perfect Execution) at $+1.30$; no spell in the pool contradicts the rule, whereas the four spells that split positioning and enhancement across different units show zero residual. Because those four also resolve as a single spell, the premium tracks target identity rather than single resolution: it is an interaction term in an otherwise additive hedonic model. A natural reading is complementarity—an enhancement placed on the unit that the same card sends into combat is guaranteed to reach the fight it creates. An alternative "atomicity" reading, that one card denies the opponent a response window between the two effects, is not supported: it would predict the same premium for the split-target spells. It is the only interaction term in the model; it applies to spell base values only, since the analogous combinations inside unit triggers and optional modes (e.g. *Pyke, Dockside Butcher* (UNL-028), *Hungry Wolf* (VEN-125)) have no independent cost evidence. With the premium booked, the five cards move from $\eta = 0.67$–0.75 to 0.94–1.00, and Perfect Execution's implied Flow premium falls from 1.5 to 0.5 C (Section V-D). With five supporting cards we treat the rule as a hypothesis. A pattern of the opposite sign—a "then draw a card" rider priced at about half of a stand-alone draw on six spells—was noted in the same review but is left unbooked pending more cards.

VI. Results: Latent Design Regularities

The following regularities emerged while fitting individual prices; each is stated with the evidence that supports it.

TABLE V
PRICES FOR EXTENDED MECHANISMS RECOVERED BY ZHIZHI (STAGE 1), IN C

Mechanism	Price	Evidence
Play-timing flexibility (ACTION, REACTION, HIDDEN, REPEAT)	0	Group comparison over 47 spells (Section VI-B)
On-play effect on a unit (resource, removal, other persistent effects)	spell price $\times 1.00$	Resource effects: 7 cards, median 1.10; removal: 5 cards, median 1.12; both ratios come from the unrounded $B(c)$; under $\lfloor B(c) \rfloor$ six integer-effect units match exactly
On-play effect realised through same-turn combat (combat trick, incl. $-X$ might this turn)	spell price $\times 0.50$	Regression -0.27 ($p = 0.057$), multiplicative fit 0.55; optional-cost anchors ≥ 0.5; range 0.5–0.75
On-play effect on a 2–3 C unit (included in printed cost)	spell price $\times 0.30$, multiplied with the above	10 cards; regression -0.63 ($p < 0.001$); multiplicative fit 0.26 with combat trick fixed at 0.5
Keywords native to a 2–3 C unit (keyword rounding)	total < 1 C: $\times 0.30$ (VISION 0.12, ASSAULT 1 / SHIELD 1 0.15); ≥ 1 C: full	9 units with < 1 C keywords give up no might for the keyword; 5 of 6 with ≥ 1 C keywords do (Section VI-E)
EMPOWER X (self-paid)	option 0; bonus at 1:1	Kinkou Lifeblade, Brutal Hunter, Steel Paws (VEN-043), Punching Poro (VEN-007) all 1.00
One free Empower granted by another effect	≈ 2.4	Median Empower cost 4 C $\times$ scenario factor ≈ 0.6; Wild Claw (VEN-089) recovers 2.36 independently
Look at opponent's hand and choose a card to discard/recycle	any 4.00; non-unit 3.00; single domain 1.00	Mindsplitter (OGN-192, 11 C, 7 might; budget-method ceiling ≈ 6.5), Sabotage (OGN-156), Decree of Strength (VEN-085)
Gain XP	0.40 per point	18 units; through-origin LS 0.38 with $\lfloor B(c) \rfloor$ (0.59 unrounded); hidden by rounding on cheap units
Spend XP	0.50 per point	2 XP buys one buff ($+1$ might)
BURN	0	Blade Twirler (VEN-002), Shadow Order Disciple (VEN-095); effects cancel
Exhaust own unit (additional cost)	-1.00	Opportunity cost: the unit forgoes one move this turn; moving a friendly unit = 1 C
Same-unit combination: reposition (move, ready, grant GANKING) and boost (this-turn might, buff, ASSAULT/SHIELD) one friendly unit in one spell	sum of parts $+1.00$	Ride the Wind, Resonating Strike, Showstopper, Vault Breaker each $+1.00$; Perfect Execution $+1.30$; different-target combinations 0 (Section V-J)
Overflow cap for single-target damage / $-X$ might	8.00	Equals unconditional kill; largest linear anchor (Final Spark); only 9 units (1.9%) have might ≥ 9
Spell without ACTION/REACTION/HIDDEN: $\pm X$ might this turn, stun	spell price $\times 0.50$	Same reasoning as combat-trick on-play effects; Onslaught (VEN-081) 4 C for $+6$ (≈ 0.67/pt) vs. Punch First 5 C for $+5$ with Action (1/pt); $-X$ might this turn takes the same factor; ready is not discounted; Assault/Shield grants, already at 0.5 per point, are not discounted again

A. The 50% Rule for Contingent Value

Across the 162 units with a repeatable trigger ("whenever" effects, moves, conquer and hold, activated abilities and auras), the median binned ratio $\eta_{(t)}$ at $t = 0$, 1, 2 and 3 triggers is 0.80, 1.20, 1.50 and 1.85 respectively, crossing 1.00 at about half a trigger (Fig. 1(a)). The designer thus prices a repeatable benefit as if it fired half a time. The same factor recurs in independent fits: DEATHKNELL at $\lambda = 0.50$; combat-trick on-play effects at 0.5 (range 0.5–0.75); scenario-dependent effects (kill a gear, free Empower) at ≈ 0.6; and some one-off conditional bonuses (*Disintegrate* (OGN-005): 3 damage $+0.5\times$ draw = 4 C for 4 C; *Shock Blast* (VEN-059): 5 C for 4 damage with a conditional 2 C reduction, $5 - 0.5 \times 2 = 4$ C)—but, as shown below, not conditional bonuses as a class. A consequence is that a "whenever" unit exceeds its cost after a single trigger, so repeated triggers are the main source of value in this class. *Stellacorn Herder* (SFD-048) (4 C, 3 might, draw 1 when it moves) has $\eta_{(t)} = 0.75, 1.25, 1.75, 2.25$ at $t = 0$–3; its implied effect price, $4 - 3 = 1$ C, is exactly half the single-trigger value. *Traveling Merchant* (OGN-185) (2 C, 2 might, discard then draw on move, net 1 C) goes from 1.00 to 1.50 after one trigger.

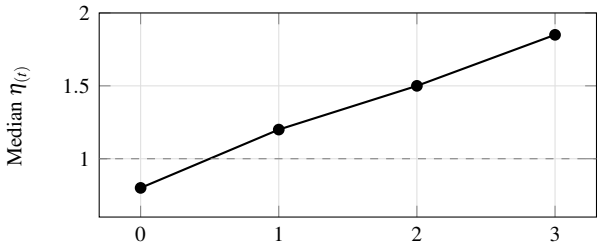
Implied activations. For each such unit the number of triggers that the printed cost pays for is $n^* = (C_{\text{tot}} - V_0)/(V_1 - V_0)$. Over the 132 units whose single trigger is worth at least 1 C (for the other 30 the increment is too small for the 1 C cost grid to resolve), n^* has median 0.500 (bootstrap 95% interval 0.354–0.500, 2000 resamples), and 78.8% of the units lie between 0 and 1 (Fig. 1(b)). The distribution is spiky—40 units sit exactly at 0, 20 at 0.5 and 15 at 1—and the upper end of the interval equals the median because of the mass at half a trigger.

Conditional triggers. Seventeen of the 132 have a trigger sentence that requires an independent condition besides the triggering event (Section III-D). Their n^* has median 0.00 (95% interval 0.00–0.26), against 0.50 (0.50–0.67) for the other 115 (Mann–Whitney $p = 0.003$): the designer charges almost nothing for a repeatable effect gated by a further condition. The p-value is descriptive: the 17 cards were tagged, by rules text, after the ranked list had been reviewed (see *Circularity* below). The AMBER score therefore books these triggers at a quarter activation, 0.5×0.5, a rule value of the same origin as the conditional-bonus factor above rather than a fitted one. *Yasuo, Windrider* (OGN-205) (third move in a turn), *Swain, Visionary* (VEN-065) (conquer after playing three kinds of card in the turn) and *Apprentice Smith* (SFD-041) (a move that also turns up

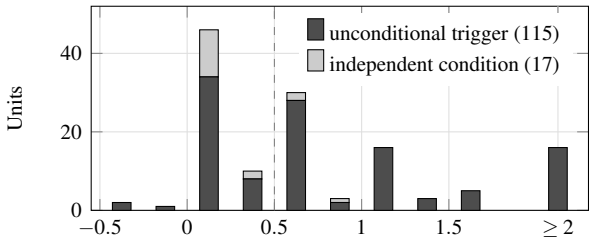
a gear) have nominal ratios 0.71, 0.80 and 1.00; booking their triggers at 0.5 would give 1.54, 1.52 and 1.50, and their AMBER scores are 1.13, 1.16 and 1.25.

Conditional bonuses. The 50% rule does not extend to one-off "if …, then …" bonuses. Among the 497 fully priced units and spells, 54 carry such a bonus—an extra effect or an upgrade contingent on a state outside the card—whose value Δ_{cond} is booked at zero in the base scenario and in full in the full-trigger reference. The 54 were selected by rules text before costs were compared; cards whose gap between the two references is explained by another convention (DEATHKNELL, FLOW, REPEAT, optional additional costs, EMPOWER, level thresholds, extra targets, conditional cost reductions, count-dependent tiers, activated abilities) were set aside. For the 46 with $\Delta_{\text{cond}} \geq 1$ C, the fraction of the conditional value that the printed cost pays for, $f^* = (C_{\text{tot}} - V_B)/\Delta_{\text{cond}}$, has median 0.08 (95% bootstrap interval 0.00–0.50, 2000 resamples), against 0.50 for n^*, and the distribution is two-humped rather than centred (Fig. 1(c)): 20 cards sit exactly at 0 (3 more below it and one at 0.17), 16 lie between 0.5 and 0.8 (8 exactly at 0.5), 3 at exactly 1 and 3 above. The designer charged nothing for *Isolate* (UNL-124) (draw a card if a lone enemy unit is at the battlefield), *Tomb-Raider Barbara* (VEN-037) (destroy an enemy gear with seven or more runes), *Dropboarder* (SFD-072) (enter ready with two or more gear) or *Poppy, Paragon* (UNL-116) (enter ready with 3 XP when the opponent is within 3 points of winning), and one half for Disintegrate, *Xin Zhao, Vigilant* (SFD-176) (enter ready with two other units in the base) and *Sudden Storm* (SFD-017) (4 instead of 2 damage against an attacker). Where the condition is one the player arranges—runes paid or a spell cast this turn, a face-down card held, or a LEGION sentence—the median is 0.67 ($n = 9$, against 0.00 for the other 37; Mann–Whitney $p = 0.052$); we record as a hypothesis, not a rule, that the designer books contingencies the player controls at about one half and contingencies on the game state at zero. The subclasses set aside behave differently again and are reported descriptively: conditional cost reductions have median f^* 0.25 ($n = 25$; Shock Blast at 0.50), level thresholds 0.50 ($n = 10$) and extra targets 0.50 ($n = 3$). The paired leave-one-out test of Section VII-D confirms that no single factor fits the class: booking Δ_{cond} at 0.5 does not improve prediction card by card, booking it in full makes it significantly worse, and the least-squares factor is 0.30 (95% interval 0.12–0.48; 0.17 without the player-controlled cards). The base scenario therefore keeps conditional bonuses at zero (Section III-D).

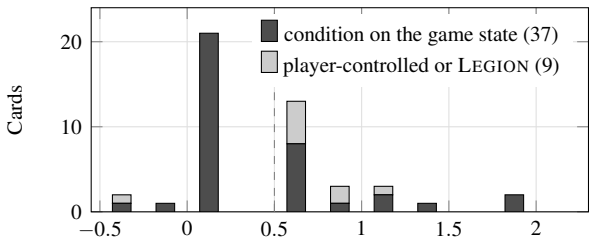
Circularity. The 0.5 factor was estimated on these 162 units, whose trigger-bin medians cross 1.00 near half a trigger; that their median AMBER score is 1.00 (0.80 nominal) is therefore a consequence of the construction, not evidence that this class is balanced, and the same holds for the smaller mean residual of Section VII-D. The 17 conditional cards were identified, by rules text, while reviewing the ranked list. The evidence for the two factors is limited to the shape of the n^* distribution (median 0.5 with an interval that excludes 0 and 1; conditional triggers concentrated at 0), its agreement with the independently



(a) Number of triggers t



(b) Implied activations n^* (132 units, bins of 0.25)



(c) Implied realisation fraction f^* (46 conditional bonuses, bins of 0.25)

Fig. 1. The 50% rule and its limit. (a) Median binned value/cost ratio $\eta_{(t)}$ of the 162 units with a repeatable trigger, by number of triggers booked (dashed line: par, 1.00); the curve crosses 1.00 near half a trigger, and the AMBER score (Section III-D) adopts this 0.5-activation baseline as its ranking convention. (b) Implied activations $n^* = (C_{\text{tot}} - V_0)/(V_1 - V_0)$ for the 132 units whose single trigger is worth at least 1 C (bins of 0.25, left-closed; the dashed line marks 0.5). The distribution is spiky: 40 units sit exactly at 0 (9 of the 17 conditional triggers among them), 20 at 0.5 and 15 at 1. (c) Implied realisation fraction $f^* = (C_{\text{tot}} - V_B)/\Delta_{\text{cond}}$ for the 46 fully priced cards whose one-off conditional bonus is worth at least 1 C (bins of 0.25, left-closed; dashed line 0.5). Unlike n^*, the mass is at zero (20 cards exactly at 0) with a second group at 0.5–0.8, most of it cards whose condition the player controls; the base scenario books these bonuses at zero (Section III-D).

fitted DEATHKNELL and combat-trick factors of 0.5, and paired leave-one-out intervals that exclude zero (Section VII-D). Their out-of-sample test on the next set, and the zero booking of conditional bonuses, are registered as hypotheses H9 and H10 in an addendum to the preregistration (Data Availability). The conditional-bonus figures share the circularity: f^* and the least-squares factor were estimated on the same 46 cards, although the list itself was fixed by rules text before costs were compared.

B. Play Timing Is Free; Choice Is Priced

For 47 spells whose effects are fully priced, grouping by timing keyword and comparing cost minus priced effect value (a positive value means the attribute is charged for) gives Table VI. Cards with timing keywords pay nothing extra; the most flexible group (REACTION) has the *lowest* mean (−0.37 C), and the

TABLE VI
EXTRA COST OF TIMING KEYWORDS OVER 47 FULLY PRICED SPELLS

Group	n	Mean (cost − value), C
No timing keyword	11	−0.09
ACTION	14	−0.14
ACTION + HIDDEN	5	−0.28
REACTION	12	−0.37
REACTION + REPEAT / HIDDEN	5	0.00

95% interval for all keyword-bearing spells is −0.61 to +0.36 C. Same-effect contrasts show no price difference either: Vengeance vs. Blast of Power, Falling Star vs. Incinerate, Progress Day vs. Downstage Dramatics. By contrast, modal cards pay ≈0.8 C above their best mode (Section V-C).

This is not a null result attributable to noise. A wider play window is a strict advantage in game-theoretic terms, and in other CCGs instant-speed effects usually carry a body or effect discount. In Riftbound the advantage is uncharged. The plausible design rationale is that REACTION and ACTION are used as domain-identity tools and to encourage interactive, second-mover play, and are deliberately kept free of a cost gate. It is a defining feature of this game's cost model: numbers buy might and effects; timing is a free qualitative tag. The one exception is a this-turn might change ($\pm X$) or stun on a spell *without* any timing keyword, which shares the unit's inability to act inside a showdown and is booked at 0.5 (only *Onslaught* (VEN-081) calibrates it; low confidence). That factor enters Table VI only through Onslaught, the one fully priced spell without a timing keyword that carries such an effect; scoring it at full price instead would move the no-keyword mean by at most 0.3 C, within the interval above.

C. Diminishing Selection Premia

PREDICT, look-N-take-1 and modal effects are all expected-maximum problems, and their premia are concave in the number of options. Moving from two to three modes adds ≈0.4 C, to four ≈0.25 C, to five ≈0.2 C; for look-N-take-1 the increments from $N = 2$ to 5 are ≈0.28, 0.17 and 0.12 C. Within the common range of 1–5 cards, PREDICT is well approximated linearly at 0.40 C per card.

D. XP Is Priced at Both Ends but Hidden by Rounding

Fitting the gain-side price over all 18 XP-gaining units (might at $\lfloor B(c) \rfloor$; repeatable gains such as HUNT and level conditions discounted by the 50% rule) gives 0.38 C per point by through-origin least squares (0.59 C without rounding); we adopt 0.40 C, close to the spend-side 0.50 C. The sample is bimodal: 12 cheap or at-par units imply ≈0, and 6 units with might below cost (*Herald of Spring* (UNL-034), *Mister Root* (UNL-127), *Scorch-claw* (UNL-016), *Mosstomper* (UNL-047), *Gustwalker* (UNL-075), *Kha'Zix, Mutating Horror* (UNL-143)) imply ≈1 C per point. The former are not unpriced: on 2–5 C units the budget slack is under 1 C and 0.4 C of XP is absorbed by rounding.

TABLE VII
RECOVERED ON-PLAY EFFECT FRACTION BY UNIT COST (25 UNITS)

	2–3 C	≥ 4 C
Combat-trick type	5 cards, mean 0.12 (Teemo, Scout; Gem Jammer; First Mate; Bubble Bot; Solari Shieldbearer)	2 cards: Field Musicians (VEN-026) 0.33, White-flame Protector 0.75
Other on-play	5 cards, mean 0.27 (Tideturner; Pit Rookie; Evershade Stalker; Lecturing Yordle; Windsinger)	13 cards, median 1.00

Level-6 and level-11 conditions show negative residuals, suggesting high thresholds are booked below 50%. Because XP has positive feedback, XP cards that reach their thresholds tend to exceed their cost, making them the natural place to look for outliers in future sets.

E. Cheap Units Get On-Play Effects Almost Free

Stratifying the 25 single-on-play-effect units by cost (Table VII) shows a sharp layer: 2–3 C units price the effect at roughly 30% of spell price regardless of type (best fit 0.26 with the combat-trick factor fixed at 0.5), while units at 4 C and above pay full price. Integer might is the mechanism—*integer-rounding absorption*: attaching a 1–4 C effect to a cheap unit removes 0 or 1 might, so most of the effect is a gift. When the same effect is sold as an optional additional cost on a cheap unit, it is priced at 0.5 or more, so the designer is not shrinking cheap bodies to pay for small effects. The factor is an average; individual cards scatter (*Scuttle Crab* (UNL-053) has 2 fewer might than cost and its on-play draw is at nearly full price). The factor applies to effects included in the printed cost; it is not applied to optional-additional-cost effects, tokens, negative cost-type effects, or single-anchor cards such as Bewitching Spirit; keywords follow the separate rule below. Applying this rule returns a cluster of 2–3 C units previously at $\eta = 1.25$–2.00 to near 1.

Keyword rounding. Keywords that a 2–3 C unit carries natively are rounded in the same way, but only when their total price is below 1 C. Of the nine such units carrying VISION (0.40 C), ASSAULT 1 or SHIELD 1 (0.50 C), seven have might equal to cost, and the two that give up one might (*Black Rose Dignitary* (UNL-152) and *Lucian, Gunslinger* (SFD-028)) do so for a Deathknell and an attack trigger; at full keyword price the seven sit at $\eta =$ 1.17–1.25, Black Rose Dignitary at 1.17 (with its Deathknell at $\lambda = 0.5$) and Lucian, whose attack trigger lies outside the base value, at 0.83. Keywords worth 1 C or more are paid for in might by five of the six 2–3 C units that carry them (e.g. *Inferna* (UNL-002), Assault 2; *Pyke, Dockside Butcher* (UNL-028), GANKING). The model therefore books sub-1 C native keywords on 2–3 C units at 0.30 of their price (VISION 0.12 C, ASSAULT 1 or SHIELD 1 0.15 C), which returns the seven units to 1.05–1.07 (Black Rose Dignitary 1.05, Lucian 0.72), and

books keywords worth 1 C or more in full; keywords gained on a condition or on EMPOWER, or granted to other units, are not rounded. *Laurent Bladekeeper* (SFD-096) (3 C, 3 might, GANKING) is the one unit that does not pay for a 1 C keyword; at full price its $\eta = 1.33$ is recorded as a genuine excess and it is not used as an anchor. The rule stops at 3 C because of the budget slack $B(c) - c$: at 2–3 C it is 0.2–0.3 C, less than the keyword's price, so the keyword exceeds the budget and is a gift; from 4 C the slack (≥ 0.4 C) pays for it, and at full keyword price the 4–6 C units sit on or under the budget line (*Jhin, Meticulous Killer* (UNL-089) at $4.40 = B(4)$, *Petty Officer* (OGN-215) at $5.50 = B(5)$), with *Jeweled Colossus* (OGN-086) 0.4 C above it. Extending the rule to 4 C and above would discount twice and raises the full model's leave-one-out RMSE from 1.545 to 1.552 (on the v5 table, on which this check was run). The boundary is the one place where the unrounded budget slope matters (Section IV-C): with a slope of 1.05 the 4 C slack would be 0.2 C and would not cover a keyword, and with 1.15 the 3 C slack of 0.45 C would cover VISION. The two groups of units behave alike (neither gives up might for the keyword) and the out-of-sample difference between stopping at 3 C and at 4 C is below resolution, so the 3 C boundary is a convention that follows from the slope 1.1 rather than a finding.

F. Flow Is Booked on the First Play

The two plays of a Flow card have different cost structures; the saved hand card (1.00 C) is realised on the Flow play, so with $F = C_{\text{tot}}$ the Flow play is the more efficient. The premium the designer books on the printed side for the keyword is small (0.2 C adopted, range 0–0.6 C; Section V-D) and vanishes once $F \geq C_{\text{tot}} + 1$.

G. Scenario Effects at About Six Tenths

Unconditional gear removal is priced from four anchors: *Brittle Steel* (VEN-003) (2 energy + 1 power = 4 C; Flow cost 2 C above printed, so $k = 0$) 4.00; *Disarming Rake* (SFD-032) (5 C unit, 2 might, may kill a gear on play) $B(5) - 2 = 3.50$; *Salvage* (OGN-224) (4 C, kill a gear and draw 1) 2.00 after the draw; *Turn to Dust* (UNL-070) (2 C, gear becomes TEMPORARY) 2.00, a lower bound because the kill is delayed. Median 2.75 C (mean 2.88); we adopt 2.75. As a magnitude check only—not a derivation—kill-a-unit (8 C) is about 1.6× the median unit total cost (5 C); the same multiple on the median gear cost (3 C) gives ≈4.8 C, and $2.75/4.8 \approx 0.57$ coincides with the free-Empower factor (median Empower cost 4 C $\times 0.6 = 2.4$ C; *Wild Claw* (VEN-089) recovers 2.36 C). A reasonable reading is that both are scenario-dependent effects—the opponent does not always have a gear or an Empower target—that the designer books below full value. The ≈0.6 factor is an empirical observation, not a consequence of the magnitude check.

H. Deathknell Booking by Set: A Hypothesis

To test whether λ varies by set, we add units whose Deathknell creates tokens (tokens at might +0.40) and *Vendetta* units that gain Deathknell on Empower (Empower cost at 1:1, other Empower benefits deducted). Table VIII gives per-set fits. The supplementary cards and their implied coefficients are *Machine Evangel* (OGN-239) $(8-4)/4.2 = 0.95$ and *Ferrous Forerunner* (SFD-021) $(9-6)/6.8 = 0.44$ (two 3-might tokens), *Carrion Dredger* (UNL-153) $(2-1)/1.4 = 0.71$, *Noxian Emissary* (VEN-128) $3/2.8 = 1.07$ (Empower 1 Order, 3 C, for two 1-might Recruit tokens) and *Baccai Witherclaw* (VEN-078) $3/4 = 0.75$ (Empower 5 C less the +2 might it also grants, for two runes). Point estimates trend upward with release order (0.53, 0.44, 0.69, 0.86; fitted line from ≈0.52 to ≈0.80), but a linear regression of λ on release order over all 10 samples has slope +0.093 per set (s.e. 0.060), Spearman $\rho = 0.44$, one-sided permutation $p \approx 0.10$; pooling *Origins+Spiritforged* (mean 0.55) against *Unleashed+Vendetta* (mean 0.74) gives a difference of 0.20 with 95% CI −0.14 to +0.54. Table VIII and the regression use Black Rose Dignitary at $\lambda = 0.43$ (Assault 1 at the keyword-rounded 0.15 C); at the full keyword price it would be 0.25, which slightly weakens the trend without changing the conclusion. None of the three tests rejects a common λ, so "later sets book Deathknell more conservatively" is recorded as a hypothesis, not a finding; the model uses $\lambda = 0.50$ for all sets. If confirmed, early cards such as Tasty Faefolk would be over-budget at the unit level and later cards such as Ruined Rex ($\eta_B = 0.84$) slightly under-priced by a uniform λ. Caveats: the late-set sample is five cards, the two *Vendetta* cards are recovered through the Empower route, and the excluded outlier Undercover Agent ($\lambda = 1.50$) would raise the *Origins* mean if included.

TABLE VIII
DEATHKNELL COEFFICIENT λ BY SET (RELEASE ORDER)

Set	n	λ (LS / median)
1. Origins	4	0.53 / 0.50
2. Spiritforged	1	0.44
3. Unleashed	3	0.69 / 0.71
4. Vendetta	2	0.86 / 0.91
All (single-Deathknell units only)	5	0.46 / 0.50
All (with token and Empower samples)	10	0.58 / 0.61

I. Ban Status Is Not Evidence of Single-Card Excess

Bans usually address combinations, and ban status is not a pricing criterion (Section IV-E). Twelve cards are banned in Standard constructed as of 24 September 2026 (nine since 31 March 2026, *Stealthy Pursuer* (OGN-177) since 24 July, *Ekko, Recurrent* (OGN-110) and *Stacked Deck* (OGN-183) since 18 September). The seven costed ones span the AMBER scale rather than its upper tail. Stacked Deck (1 C, look-3-take-1) scores 2.83—the same value as the unbanned *Lightning Rush* (VEN-156), which trashes rather than recycles the other two cards; *Called Shot* (SFD-122) 1.28; *Draven, Vanquisher* (SFD-020) 1.11 (1.00 nominal); *Scrapheap* (OGN-182) 1.00 (gear; its second draw, on discard or kill, is conditional); *Fight or Flight* (OGN-168) 1.00 (2 C to move a unit from a battlefield to its base, with HIDDEN

and ACTION at zero); Stealthy Pursuer 0.75 (0.67 nominal: 6 C for 4 might and a move-along trigger); and Ekko, Recurrent is deferred because its Deathknell readies all runes (0.71 on the priced part, 5 might for 7 C). Of the five banned battlefields, *Aspirant's Climb* (OGN-276) changes a rule, *The Arena's Greatest* (OGN-290) and *Obelisk of Power* (OGN-284) are symmetric (a point, or a rune, to each player), *Reaver's Row* (OGN-285) yields one friendly move per defence and *The Dreaming Tree* (OGN-292) one draw per turn to whoever meets its condition. What the bans have in common—loops, deck-thinning consistency, tempo and first-player effects—lies in the value-fairness layer that the score does not measure. Three of the banned spells and two of the battlefields had been set aside under an earlier version of the deferral rules and remain outside the statistics so that the pre-registered frozen model is unchanged (Table I); the other seven carry no such flag and are priced like any other card.

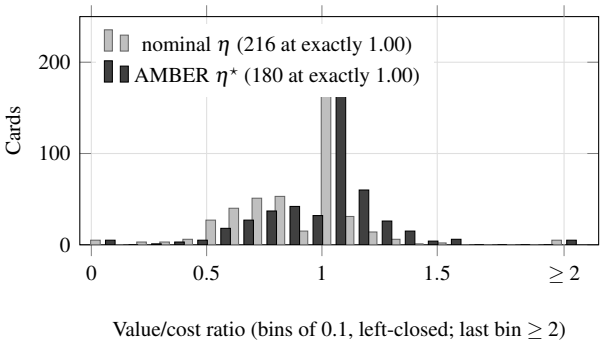


Fig. 2. Distribution of the nominal ratio and the AMBER score over the 497 fully priced units and spells. The bin $[1.0, 1.1)$ holds the cards at exactly 1.00, 146 of which are anchors; booking repeatable triggers at half an activation moves 55 units out of that bin to the right and 19 into it, and shifts the left tail of the 162 trigger units towards par.

VII. VALIDATION

A. Hold-Out Predictions

The model's credibility rests mainly on predicting cards that were not used to calibrate the relevant parameter (Table IX); each row is read as prediction interval against observed value at the model's ≈0.3 C resolution. The first row is a consistency check rather than a hold-out: the adopted PREDICT dispersion was fitted jointly on Clairvoyance, Apprentice Mage and the discard price, and under it the order-statistics model implies 0.83–1.04 C for the lowest card in hand against the 1.00 C recovered from three cards, so the discard price and the selection premia are mutually consistent rather than one predicting the other. The zero intercept of the damage pair predicts that draw cards are priced through the origin, and they are. The power exchange rate from three spell equations agrees with the price of channelling a rune from two other cards, on the reading that a rune is a unit of power; this too is a consistency check between separately anchored prices. The free-Empower value from the Empower cost distribution matches Wild Claw. The opponent-chooses variant of the modal model predicts Shakedown to within 0.16 C. The Tactical Retreat bundle (remove damage, recall to base, exhaust) at ≈2 C matches the cost-minus-might of *Soraka, Wanderer* (SFD-173), where the same bundle is a trigger. The value of playing directly to a battlefield, calibrated on two units, matches a third unit at 1.00 C (a fourth at 0.60) and the yield of the *Dragon Roost* (VEN-157) battlefield valued as one friendly move.

B. Distribution of the AMBER Score

Over the 497 fully priced units and spells, the AMBER score $\eta^\star$ has mean 0.972, median 1.00 and interquartile range 0.86–1.07; 62 cards exceed 1.17, 11 reach 1.50, and 36.2% sit exactly at 1.00. The nominal ratio η has mean 0.911, the same median and interquartile range 0.75–1.00, with 32 cards above 1.17, 7 at or above 1.50 and 43.5% at 1.00. The difference is confined to the 162 units with repeatable triggers (spells are unchanged): the 55 of them whose base value equals their cost move above 1.00, and 19 others move up to it. Two qualifications apply. Anchor cards score 1.00 by construction; excluding the 146 of them leaves 351 cards with median 1.00, interquartile range 0.80–1.09 and 30.5% exactly at 1.00 (nominal: median 1.00, 0.71–1.00). And for the 162 trigger units the return to a median of 1.00 is partly a construction of the 0.5 factor (Section VI-A, *Circularity*). Fig. 2 shows both distributions. Under either score the recovered table reproduces the designer's pricing for the central mass of the pool, with a left tail of conservatively priced cards and a short right tail (Section VII-F).

C. Ablation

The mechanism models were introduced incrementally; Table X measures what each adds. We compare four nested models—A, linear additive core only ((4), (5), (6)); B, A + mental accounting of hand cards (Section IV-D, closing draw/discard/Flow); C, B + order-statistics selection premia (Section IV-F); D, C + cost-controlled on-play adjustments (Section V-H)—and report RMSE, MAE and R^2 of V against C_{tot}, with residual $r_i = V_i - C_{\text{tot},i}$ on base value, over the 497 fully priced units and spells. D is the full model, and its values are those of the current value table; it also applies keyword rounding (Section VI-E), which is part of layer D. Removing a layer applies the counterfactual rule of the smaller model: A and B credit PREDICT at 0.60 C per card (Clairvoyance's 7 C less two draws, over five cards), price look-*N*-take-1 as a single draw and modal effects at their best mode without premium (worst mode when the opponent chooses); A books no Flow keyword premium; A–C price units' on-play effects and keywords at full spell price. Discard is −1.00 C, the non-showdown factor on spells is 0.50, and the same-target complementarity premium (Section V-J) is booked in every model, since none of these is an ablated mechanism. Every price whose calibration cards use a removed mechanism is re-estimated on the same cards with the same statistic; in practice only the token premium moves (0.40 to 0.50 C under A, because two of its thirteen calibration spells carry Flow).

Under leave-one-out cross-validation [28], each card is held out, every anchor-derived price is re-estimated from the remain-

TABLE IX
CROSS-PREDICTIONS BETWEEN SEPARATELY CALIBRATED PARAMETERS (ROWS MARKED C: CONSISTENCY CHECKS, NOT HOLD-OUTS)

Quantity predicted	Calibrated from	Prediction	Observed
(C) Discard the lowest-quality card in a hand (own -1.00; opponent's, their choice, $+1.00$)	Order-statistics model with the jointly fitted dispersion	0.83–1.04 C	1.00 C (Lunar Boon, Chemtech Enforcer; Bewitching Spirit for the opponent's hand)
Zero hand-rent intercept (playing is rent-free)	Damage pair Incinerate (2 C, deal 2) / Final Spark (8 C, deal 8): 1 C per point, intercept 0	Other effect lines pass through the origin	Draw: Consult the Past (4 C, draw 2), Progress Day (8 C, draw 4) give 2 C per card, intercept 0
(C) 1 power	Three spell equations	2 C	Channel 1 exhausted rune = 2 C (Mobilize, Catalyst of Aeons), reading a rune as a unit of power
One free Empower	Median Empower cost $\times 0.6$	2.4 C	Wild Claw recovers 2.36 C
Ready 2 C, buff 1 C, return-from-trash 2 C	Individual anchors (Upstage Comedy, Wallop; Morbid Return)	base $\eta = 1.00$	Call to Glory (OGN-207), Confront (OGN-129), Rally the Troops (SFD-166), Here to Help (SFD-111), Undying Loyalty (UNL-168) all 1.00
Opponent chooses the worse mode	Modal model (expected minimum)	3.84 C	Shakedown printed 4 C
Remove damage + recall + exhaust bundle	Zhonya's Hourglass (OGN-077), Tactical Retreat (UNL-175)	$\approx$2 C	Soraka, Wanderer: cost $-$ might = 2 C for the same bundle as a trigger
May be played directly to a battlefield	Sneaky Deckhand, Rengar, Pouncing (anchors)	$+1$ C	Ocean Drake 1.00 C, Sai Scout 0.60 (units not used as anchors); Dragon Roost battlefield yield, valued as one friendly move, 1 C

ing cards with the same statistic and chaining (a price adopted as a rounded statistic is shifted by the change in that statistic; the token premium and the Flow coefficient calibrate each other and are iterated to a fixed point), and the held-out card is then predicted. Sixteen cards cannot be predicted because a price they use is identified by that card alone—for example *Rune Prison* (OGN-050) for an unrestricted stun, Clairvoyance for PREDICT, Onslaught for the non-showdown factor—or, for *Guerilla Warfare* (OGN-264), by a card-specific value; they are dropped from all four models, leaving 481 cards (354 units, 127 spells). Two other card-specific values are replaced by the nearest global price when their card is held out: *Sacrifice* (UNL-173) takes the ally-sacrifice price (-2 C) and *Turn to Dust* (UNL-070) the median of the other three gear-destruction cards.

Only layer D reduces error materially. Over all cards it lowers the sum of squared residuals (SSE) by 49.7 relative to C (RMSE 1.537 to 1.504, R^2 0.665 to 0.679); of the 26 units it changes, 22 move closer to their cost, 3 move away and 1 is unchanged. The gain survives cross-validation almost intact (SSE -49.3; RMSE 1.574 to 1.541, R^2 0.650 to 0.665), with 25 of the 29 changed units improving, so the low-cost and combat-trick factors are not artefacts of their own calibration cards. D leaves spells unchanged by construction.

Layers B and C change few base values, and their effect is at the resolution limit. C alters 21 cards and lowers SSE by 1.2 in-sample (1.5 on the cross-validated subset, and the same out of sample), but its MAE is marginally higher than B's (0.9103 against 0.9099) and it fits spells slightly worse, mainly because the look-3-take-1 premium raises the residual of *Lightning Rush* (VEN-156) from $+1.00$ to $+1.83$ C. B alters 16 cards and lowers SSE by 0.5 in-sample; on the 481-card subset the change is -0.2 in-sample and -0.1 out of sample (spells $+0.1$ in both), i.e. within resolution in either direction. The Flow coefficient rests on four calibration spells, and its leave-one-out cost is the variance of that estimate rather than evidence against the keyword: with k fixed at 0.2 the B layer's out-of-sample SSE is lower than with $k \equiv 0$ (by 0.6 before the complementarity premium of Section V-J was booked), and no estimation rule for k predicts held-out Flow cards better than any other (Section VII-E). Under cross-validation the spells are fitted about equally by all four models (RMSE 1.489; MAE 0.842–0.847, lowest under A). The small size of B and C is expected: discard is priced identically in every model, only five Flow cards have $p = 0$, and selection premia enter 21 base values.

Because 436 of the 497 cards take the same value under all four models, full-sample metrics dilute any layer that touches few cards. Table XI therefore repeats the comparison on the cards each layer changes. On its own 26 cards (29 under leave-one-out), D lowers RMSE from 1.46 to 0.46 C in-sample and from 1.38 to 0.45 C out of sample; both 90% bootstrap intervals exclude zero, and 22 of 26 (25 of 29) cards move closer to their cost. B and C stay indistinguishable from the layer below even where they act. On B's 16 cards RMSE falls by 0.028 C in-sample and 0.003 C out of sample; on C's 21 (20) cards by 0.027 (0.035) C. Both intervals include zero in both evaluations, nearly as many cards get worse as better (9/7 and 8/7 for B; 12/9 and 12/8 for C), and the paired signed-rank tests give $p \geq 0.61$. The small full-sample effect of B and C is therefore not only dilution: on the cards they change, their corrections are of the order of the per-card noise. With 16–29 cards per subset these results are descriptive and are not read as significance tests of the mechanisms.

Printed costs move in 1 C steps, so a layer could be informative without lowering squared error if it moved cards onto their printed integer. We therefore also counted, on the same subsets, the cards whose value rounds to the printed cost ($|V - C_{\text{tot}}| < 0.5$) before and after each layer, with an exact McNemar test (one-sided, for an increase). The count is a valid check for D, the

TABLE X
ABLATION: IN-SAMPLE AND LEAVE-ONE-OUT ERROR OF V AGAINST C_{tot}

Model	In-sample RMSE (C)	In-sample MAE (C)	In-sample R^2	Leave-one-out RMSE (C)	Leave-one-out MAE (C)	Leave-one-out R^2
All fully priced cards (in-sample $n = 497$; LOO $n = 481$)						
A: linear core	1.538	0.910	0.664	1.575	0.957	0.650
B: A + hand-card mental accounting	1.538	0.910	0.665	1.575	0.957	0.650
C: B + order-statistics premia	1.537	0.910	0.665	1.574	0.957	0.650
D: C + on-play cost adjustments (full)	1.504	0.871	0.679	1.541	0.916	0.665
Units (in-sample $n = 359$; LOO $n = 354$)						
A: linear core	1.592	0.980	0.663	1.605	0.998	0.656
B: A + hand-card mental accounting	1.592	0.979	0.663	1.605	0.998	0.656
C: B + order-statistics premia	1.591	0.977	0.663	1.603	0.996	0.657
D: C + on-play cost adjustments (full)	1.547	0.923	0.682	1.559	0.941	0.676
Spells (in-sample $n = 138$; LOO $n = 127$)						
A: linear core	1.388	0.728	0.556	1.489	0.842	0.506
B: A + hand-card mental accounting	1.387	0.730	0.556	1.489	0.844	0.506
C: B + order-statistics premia	1.388	0.736	0.556	1.489	0.847	0.506
D: C + on-play cost adjustments (full)	1.388	0.736	0.556	1.489	0.847	0.506

Residual $r_i = V_i - C_{\text{tot},i}$ on nominal (base) value; $R^2 = 1 - \sum r_i^2 / \sum (C_{\text{tot},i} - \bar{C}_{\text{tot}})^2$. Leave-one-out: each card is held out and every anchor-derived price is re-estimated without it under the same layer (including chained re-estimates), then the card is predicted. Sixteen cards whose price is identified only by the card itself (single-anchor prices and one card-specific value) cannot be predicted and are dropped from all four layers; in-sample error on the same 481 cards is RMSE 1.563 / 1.563 / 1.562 / 1.528 and MAE 0.938 / 0.938 / 0.938 / 0.897 for A / B / C / D. Values are those of value table v6 (the same-target complementarity premium of Section V-J is booked in every layer). Known effect of D over C on units (nominal ratio): $\eta \geq 1.5$ from 8 to 1, $\eta > 1.17$ from 29 to 13 (with keyword rounding); median and quartiles unchanged (1.00; 0.75–1.00).

TABLE XI
ABLATION ON THE CARDS EACH LAYER CHANGES

Layer	Eval.	n	RMSE (C)	ΔRMSE [90% CI]	$+/-/=$	p
A→B	In	16	0.631→0.604	−0.028 [−0.07, +0.03]	9/7/0	0.81
	LOO	16	0.598→0.594	−0.003 [−0.03, +0.03]	8/7/1	0.95
B→C	In	21	1.057→1.030	−0.027 [−0.15, +0.11]	12/9/0	0.82
	LOO	20	1.083→1.048	−0.035 [−0.16, +0.10]	12/8/0	0.61
C→D	In	26	1.456→0.455	−1.001 [−1.39, −0.59]	22/3/1	<0.001
	LOO	29	1.380→0.453	−0.927 [−1.30, −0.53]	25/3/1	<0.001

Each subset holds the cards whose value differs between the two layers under that evaluation (In: in-sample, 497 cards; LOO: leave-one-out, among the 481 predictable cards). RMSE is of $r_i = V_i - C_{\text{tot},i}$ under the lower and the upper layer. $+/-/=$: cards whose $|r_i|$ decreases, increases or is unchanged. CI: percentile bootstrap over cards (10,000 resamples). p: two-sided exact Wilcoxon signed-rank test on paired $|r_i|$ (zero differences dropped, mid-ranks for ties).

positive control: 5 of its 26 cards hit under C and 18 under D (15 gained, 2 lost; $p = 0.001$; 8 to 21 of 29 under leave-one-out, $p = 0.001$). For C the direction is the same but the difference is not significant: 5 of 21 cards hit under B and 8 under C (5 gained, 2 lost; $p = 0.23$; 4 to 8 of 20 out of sample, $p = 0.11$). For B the count is uninformative: 10 of its 16 cards sit exactly ± 0.50 C from their cost under A, B moves them by 0.1 C, and whether that counts as a gain depends entirely on the tie convention (7 gained under a strict inequality, none under an inclusive one). With 16–21 affected cards the test has little power, and it does not change the reading above: whether the designer uses the hand-card and selection rules cannot be decided at the current sample size and 1 C cost granularity; we retain both layers for interpretability and because their cross-checks hold (Table IX).

The overall level of error comes from what no layer addresses. Base values exclude conditional and triggered effects, so residuals are negative on average (-0.57 to -0.62 C in-sample, -0.58 to -0.63 C out of sample) and in-sample R^2 stays between 0.66 and 0.68 in every model. Cross-validation raises RMSE by 0.012–0.013 C in every model, mostly through Sacrifice ($+3.00$ C when held out) and the four gear-destruction cards. The ablation, like the preregistered benchmark, uses base values; booking repeatable triggers as the AMBER score does removes part of this bias (Section VII-D).

D. Trigger Booking as a Paired Test

Table XII tests the trigger factors of Section III-D on the full model D under leave-one-out and, in its last two rows, the booking of one-off conditional bonuses (Section VI-A). Each trigger version adds $k(V_1 - V_0)$ to the held-out prediction of the in-scope units: *nominal* ($k = 0$), *uniform* ($k = 0.5$ for all 162 units), *baseline* (uniform, with $k = 0.25$ for the 17 conditional triggers; the AMBER score), and two versions with a coefficient per trigger type (the leave-one-out median of n^* within the type), without and with the conditional halving. Each pair is compared on the squared errors of the cards whose k differs, by a paired bootstrap over cards (2000 resamples), a sign-flip permutation test of the mean difference (10,000) and the Wilcoxon signed-rank test (exact below $n = 25$), with seed 20260924. Leave-one-out

re-estimates the effect prices in V but not the factors themselves (removing any one card leaves the median n^* at 0.5), so the table asks whether the fixed factors improve prediction card by card; it is not an out-of-sample test of the factors.

Both steps of the baseline convention improve prediction card by card. Uniform booking brings 95 of the 162 units closer to their cost (66 further) and lowers the leave-one-out RMSE over all 481 cards from 1.541 to 1.438 C; the conditional halving brings 14 of its 17 cards closer and lowers it to 1.400 C (in-sample 1.504 to 1.365 C, mean residual -0.62 to -0.29 C). Both intervals exclude zero on the changed cards and on all 481 cards, and the three tests agree. The conditional gain is concentrated: four high-increment cards (Swain, Yasuo, *Shen, Leader of the Kinkou Order* (VEN-138) and *Tryndamere, Barbarian* (OGN-034)) account for 90% of its reduction in squared error, and on the 17 cards a factor of 0.25 predicts better than 0 (RMSE 0.818 against 1.471 C). Per-type coefficients are not adopted (Section VII-E): against the uniform and baseline versions their intervals include zero, and on the 128 unconditional cards they change, they predict worse than the uniform factor.

The same test on one-off conditional bonuses (last two rows) finds no factor to adopt. Booking Δ_{cond} at 0.5 brings 23 of the 53 cards closer to their cost and 30 further, with intervals that include zero on the changed cards and on all 481; booking it in full is significantly worse than 0.5 (8 closer, 43 further). The grid minimum lies at 0.25 (ΔRMSE -0.212 C, interval -0.391 to $+0.008$; p_{perm} 0.08, p_{W} 0.06), but four high-Δ_{cond} cards (*Solari Chief* (OGN-225), *Siphoning Strike* (VEN-146), *Master Yi, Meditative* (OGS-004), Poppy, Paragon) account for 72% of its reduction in squared error while 29 of the 53 cards move away from their cost, and the three tests disagree; it is not adopted. By set, a 0.5 booking helps on *Origins* and *Spiritforged* cards and hurts on *Unleashed* and *Vendetta* (median f^* 0.50, 0.25, 0.00 and 0.00): the opposite direction to the upward drift of λ in Section VI-H, and, like it, a per-set description rather than a rule. The AMBER score keeps $k_{\text{cond}} = 0$.

E. Parameter Review

After the ablation, every price in the table was re-examined against the cards that use it. For each of the 37 multi-anchor or layer parameters we grouped the fully priced cards whose formula contains it, tested whether the clean cards among them (base value equal to full-trigger value, 199 cards) show a one-sided residual in-sample and under leave-one-out, and recorded how far the parameter moves when any one of its anchors is held out; parameters flagged by these screens were then examined within their own anchor set, comparing the leave-one-out error of alternative estimation rules and computing exact bootstrap intervals. The 15 prices identified by a single card were only cross-checked against a second card, never adjusted from residuals. A change was to be adopted only if the new value followed from the parameter's own anchors and comparable cards under the same statistic, differed from the current value by more than the 0.3 C resolution, and did not predict held-out anchors worse than the current value (or its interval excluded the current value).

No parameter met these conditions. The Flow coefficient is the case discussed above. Two screens that at first suggested changes were resolved by the might budget: the five 4–6 C units carrying a keyword worth less than 1 C (Section VI-E) give up no might for it, as the 2–3 C units do, but their keyword fits inside the budget slack $B(c) - c$, so extending keyword rounding to them would double-discount and worsens out-of-sample error (leave-one-out RMSE 1.545 to 1.552 on the v5 table, on which the check was run; Section VI-E notes that the boundary itself rests on the budget slope); and the $1.10\times$ multiplier recovered for on-play effects is the fractional slack under whole-card rounding (Section V-H). The prices of moving a friendly unit (1.00 C) and readying a unit (2.00 C) are each contradicted by a cluster of spells at $+1$ C, but the cluster is the same-target complementarity of Section V-J, not a mis-set price: any single value leaves four cards off by ± 1 C, while raising the ready price to 3 C would lower the full-sample cross-validated SSE by 10 only by lifting a set of expensive units that sit below the curve for unrelated reasons, at the cost of a 1 C error on both single-effect anchors. The gear-destruction price (2.75 C) rests on four anchors with a bimodal spread (2.0, 2.0, 3.5, 4.0; exact bootstrap 90% interval of the median 2.0–4.0), and the mean (2.875 C) is more stable under leave-one-out, but the difference is below resolution. Among single-anchor prices, the second card agrees for recurring a unit (*Cemetery Attendant* (OGN-165), *Undying Loyalty* (UNL-168)), preventing the next damage (*Tactical Retreat* (UNL-175)) and playing to an enemy battlefield (*Rengar, Trophy Hunter* (VEN-179)); it disagrees, without a way to decide, for moving an enemy to its base (*Maddened Marauder* (OGN-191) implies 1 C against Isolate's 2 C), an unrestricted stun (*Back Off* (UNL-042) implies at most 3 C, *Zenith Blade* (OGN-262) at least 6 C) and sacrificing an ally (*Sacrifice* (UNL-173) implies -5 C against Cruel Patron's -2 C); these are listed in Table XIV. Two methodological points follow. Parameters must not be tuned on the pooled residual of all cards that use them, because that residual is dominated by unpriced conditional effects and by cards that are simply weak; and unit residuals must be read against the right target—$\lfloor B(c) \rfloor$ for might, $B(c)$ for effects—since the difference between the two is exactly the size of a sub-1 C effect. The same review procedure, applied to a per-trigger-type coefficient for repeatable triggers (the leave-one-out median of n^* within each trigger type), found no stable gain: it does not predict held-out cards better than the uniform factor (ΔRMSE -0.010 C on 481 cards, 95% interval -0.060 to $+0.037$), and the low medians of move and conquer/hold triggers (0.25 and 0.27) rise to 0.43 and 0.45 once the 17 conditional triggers are removed (Kruskal–Wallis p from 0.027 to 0.13), so the apparent type differences are mostly the conditional effect (Section VII-D).

F. Highest-Ratio Cards and Outliers

Table XIII lists the 15 highest AMBER scores among fully priced units and spells, with the nominal ratio alongside (ranks 4–5 tie at 2.00, 7–11 at 1.50 and 12–13 at 1.47). Eight are spells, seven of them this-turn might changes, and their scores equal their nominal ratios. The seven units enter through their repeatable

TABLE XII
TRIGGER AND CONDITIONAL-BONUS BOOKING: PAIRED LEAVE-ONE-OUT COMPARISON UNDER MODEL D

Comparison	n	$+/-/=$	Changed cards		All 481 cards	p_{perm} / p_{W}
			RMSE (C)	ΔRMSE [95%]	ΔRMSE [95%]	
Uniform 0.5 vs. nominal	162	95/66/1	1.793→1.518	−0.275 [−0.435, −0.102]	−0.103 [−0.171, −0.036]	0.002 / <0.001
Baseline vs. uniform	17	14/3/0	1.926→0.818	−1.108 [−1.751, −0.368]	−0.038 [−0.080, −0.008]	0.023 / 0.013
By type vs. uniform	139	71/66/2	1.395→1.361	−0.035 [−0.206, +0.132]	−0.010 [−0.060, +0.037]	0.70 / 0.94
By type + conditional vs. baseline	139	68/69/2	1.252→1.321	+0.069 [−0.062, +0.207]	+0.018 [−0.018, +0.057]	0.36 / 0.46
Conditional bonus at 0.5 vs. baseline (0)	53	23/30/0	1.316→1.209	−0.107 [−0.513, +0.336]	−0.011 [−0.055, +0.037]	0.67 / 0.61
Conditional bonus at 1.0 vs. 0.5	53	8/43/2	1.209→2.053	+0.845 [+0.534, +1.128]	+0.105 [+0.049, +0.180]	<0.001 / <0.001

Squared error of $V_{\text{LOO}} + k(V_1 - V_0)$ against C_{tot}. n: cards whose k differs between the two versions; $+/-/=$: cards closer to, further from, or as far from their cost under the first version. Leave-one-out RMSE over all 481 predictable cards: nominal 1.541, uniform 1.438, baseline 1.400, by type 1.429, by type + conditional 1.418 C. Intervals: paired percentile bootstrap, 2000 resamples. p_{perm}: sign-flip permutation, 10,000; p_{W}: Wilcoxon signed-rank, exact below $n = 25$ (baseline vs. uniform), tie-corrected normal approximation otherwise. Last two rows: $V_{\text{LOO}} + k(V_1 - V_0) + k_{\text{cond}}\Delta_{\text{cond}}$ on the 53 predictable cards with a one-off conditional bonus (Section VI-A), triggers at the baseline factors; seed 20260925. Leave-one-out RMSE over all 481 cards at $k_{\text{cond}} = 0$, 0.25, 0.5, 0.75 and 1: 1.400, 1.380, 1.390, 1.429, 1.495 C.

triggers; ranked by the nominal ratio, the list instead ended with four spells and *Laurent Bladekeeper* (SFD-096) at 1.33 and *Tideturner* (OGN-199) at 1.30. Two of the four units at 1.50 or above, *Akali, Deadly Weapon* (VEN-021) and *Ambessa, Respected and Feared* (VEN-136), have trigger tiers that presuppose they are EMPOWERED; their score uses the printed cost alone as denominator, a convention still under review. The two *Decree* spells owe their ratio to the 0.5 trait-restriction factor and are live only against one domain. Moonlight Affliction (−10, capped at 8) is 1.14; Onslaught (no timing keyword, +6 at the non-showdown discount) 0.80; *Back to Back* (OGN-206) (second target as conditional value) 0.67.

Several individually analysed cards illustrate the model. *Irresistible Faefolk* (UNL-112) has $\eta_{(1)} = 1.50$ and $\eta_{(3)} = 3.50$, against medians of 1.20 and 1.85 over the 162 trigger units (Fig. 1(a)); its nominal ratio agrees with its reputation among players. Tasty Faefolk (7 C, 6 might; 7 C should carry 8 might) trades 2 might for a Deathknell worth 6 C (two runes + draw 1), booked at 3 C: $\eta = 9/7 = 1.29$, $\eta_B = 1.10$, $\lambda = 0.33$. *Astral Heron* (VEN-044) (7 C, 7 might; cost reduction of 2 energy and 2 power on the next card each turn while at a battlefield, up to 6 C per trigger) has an implied effect price of only 1 C, 0.17 of the single-trigger value and far below the 3 C the 50% rule would imply; its $\eta_{(t)}$ is 1.00, 1.86, 2.71, 3.57 at $t = 0$–3. Under the baseline convention of Section III-D, Astral Heron scores 1.43, Irresistible Faefolk 1.00 (nominal 0.50), Stellacorn Herder 1.00 (nominal 0.75) and Traveling Merchant 1.25. Weakly priced cards include *Overt Operation* (OGN-153) ($\eta = 0.44$ with two buffed allies), *Rocket Barrage* (SFD-077) (0.72, about 1.5 C above the modal model's premium), *Shadow Dash* (VEN-148) and *Twilight Step* (VEN-105) (0.50 and 0.38; restricted moves, Flow costs 5 C and 2 C above printed), Brittle Steel (0.69; its 4.00 C gear-kill anchor exceeds the adopted 2.75) and Undercover Agent (0.86; $\lambda = 1.50$, excluded from the Deathknell fit). Perfect Execution, listed as weak in the first version of the table ($\eta = 0.74$), is at 0.94 once its ready-plus-Assault combination is priced (Section V-J). On the other side, the parameter review singles out *Wages of Pain* (SFD-070) (3 C for 3 damage plus a gold token; the token is effectively free), Lightning Rush (1 C for look-3-take-1, residual +1.83 C), *Decree of Focus* (VEN-040) and *Decree of Insight* (VEN-061) (1 C Decrees whose trait restriction is effectively worth 0.2–0.25 rather than 0.5, within the 1 C minimum-cost rounding) and *Jeweled Colossus* (OGN-086) (5 might with PREDICT and SHIELD 1 at 5 C, 0.4 C over the budget slack), and at 8 C *Zephyr Sage* (OGS-005), *Vanguard Attendant* (OGS-016) and *Garen, Rugged* (OGS-007), each two might below $\lfloor B(8) \rfloor$.

VIII. DISCUSSION

A. What the Designer's Ledger Looks Like

Read together, the recovered prices describe a costing logic with a few strong commitments: a fixed 2:1 exchange between power and energy; a linear price for damage and might up to a cap that equals the unconditional-kill price; a ledger in which hand cards are an asset with a reserve price but the act of playing is rent-free; a discount of one half for repeatable and player-controlled contingencies, and nothing at all for a one-off condition on the state of the game; free play-timing; and integer rounding that gives cheap units their effects nearly for free. These are the kinds of rules a design team could plausibly maintain in a spreadsheet, which is itself weak evidence that the inverse problem has recovered something real rather than an artefact of the fit.

B. Points Are Not a Linear Resource

The price of scoring one point (11.50 C) is reproducible across six cards, but it should be read as a design-side safety threshold: the printed cost that prevents point-generating combinations from ending games too quickly. Points are the terminal objective, whereas energy, power and might are process resources; the marginal value of a point is strongly non-linear over the course of a game (an early point can cost the tempo needed to survive, a match point is decisive). The 11.50 C figure is therefore appropriate for unit-level pricing and for judging whether a point-scoring card is over- or under-budget, but it must not be used as a linear

TABLE XIII
TOP 15 AMBER SCORES AMONG FULLY PRICED UNITS AND SPELLS

#	Card	Type, cost	Nom. (η)	AMBER ($\eta^\star$)	Composition
1	Stupefy (OGN-095)	Spell 1 C	3.00	3.00	Reaction: -1 might (1) + draw 1 (2)
2	Lightning Rush (VEN-156)	Spell 1 C	2.83	2.83	Look 3, take 1 (2.83); rest to trash
3	Decree of Insight (VEN-061)	Spell 1 C	2.50	2.50	-5 might to a Body enemy (5×0.5)
4	Discipline (OGN-058)	Spell 2 C	2.00	2.00	Reaction: +2 might (2) + draw 1 (2)
4	Decree of Focus (VEN-040)	Spell 1 C	2.00	2.00	+4 might vs. a Fury enemy (4×0.5)
6	Trevor Snoozebottom (UNL-048)	Unit 3 C, 3 might	1.05	1.51	Base 3.15 $+0.5\times$ hold: play a 3-might Sprite (2.75)
7	Akali, Deadly Weapon (VEN-021)†	Unit 3 C, 3 might	1.00	1.50	3 might $+0.5\times$ move: damage (3)
7	Ambessa, Respected and Feared (VEN-136)†	Unit 5 C, 5 might	1.00	1.50	5 might $+0.5\times$ attack: kill a weaker enemy (5)
7	Nami, Headstrong (UNL-052)	Unit 3 C, 3 might	1.00	1.50	3 might $+0.5\times$ hold: ready and buff the next unit (3)
7	Cleave (OGN-004)	Spell 1 C	1.50	1.50	Action: Assault 3 this turn (1.5)
7	Grim Resolve (UNL-095)	Spell 2 C	1.50	1.50	Action: +3 might (3); XP on win is conditional
12	Eclipse (UNL-063)	Spell 3 C	1.47	1.47	Reaction: -4 might (4) + Predict (0.4)
12	Draven, Audacious (SFD-148)	Unit 8 C, 6 might	0.75	1.47	6 might $+0.5\times$ first combat win each turn: 1 point (11.5)
14	Ezreal, Prodigy (SFD-149)	Unit 5 C, 3 might	1.20	1.44	Base 6.00 (discard 1, draw 2) $+0.5\times$ cheaper optional cost (2.4)
15	Astral Heron (VEN-044)	Unit 7 C, 7 might	1.00	1.43	7 might $+0.5\times$ cost reduction on the next card (6)

Nom.: nominal ratio (triggers at zero); AMBER: triggers at 0.5 activations (0.25 under an independent condition; none of these 15). †Trigger tiers presuppose the unit is EMPOWERED; denominator is the printed cost only (under review).

exchange rate in live play decisions, where the value of a point depends on the score, the board and the remaining resources. The in-game use of the model described in Section I is thus confined to trades among process resources.

C. Transfer to Other Card Games

ZHIZHI needs a cost expressible in one numéraire, cards that isolate single effects, and a vanilla-stat curve. MTG (card advantage, scry/impulse draw, vanilla creature curves), *Lorcana* and the *Pokémon* TCG have these features, and the three mechanism models—mental accounting of hand stock, order-statistics premia for selection, and integer-rounding absorption on cheap bodies—correspond to structures they share. We expect the framework to transfer to such games, where it would yield a game-specific AMBER price table: identify exact anchor pairs first, fit global parameters only where no pair exists, keep confidence tiers tied to independent evidence, and validate by cross-prediction between mechanisms. We have not tested this. What changes across games is the exchange rate between resources, the budget curve, and which contingencies are booked at one half and which at zero.

D. Use in Procedural Content Generation

A price table with provenance is a natural constraint for card generation. Sequence models can already produce syntactically plausible cards [20], and learned cost recommenders can score them [2]; an explicit table explains why a candidate is off-budget and can be edited by a designer when a rule is changed deliberately. In procedural card generation the AMBER score can serve as a soft screen rather than a hard constraint: flag candidates outside the empirical 0.86–1.17 band for designer review and check their trigger tiers. Because the AMBER score already books half an activation, "whenever" units are no longer penalised for the triggered value that the nominal ratio leaves out: their median is 1.00, against 0.80 nominal (Section VI-A). The band is an untested empirical range (Section VIII-E). The score also supplies a fitness signal for automatic balancing [13, 16] that does not require game simulation.

E. Limitations

Scope. The model addresses unit fairness only; apart from the single complementarity premium of Section V-J it ignores synergy, and it ignores tempo and metagame; η is not a strength rating; and the fully priced sample is skewed towards cheap, common, single-domain units (Table I(b)). *Reference point.* $\eta = 1.00$ is a unit of measurement; the empirical "normal" band of 0.86–1.17 is a working reference that has not been tested formally. *Trigger booking.* The 0.5 and 0.25 booking factors are conventions estimated on the same pool; their out-of-sample test is preregistered as H9 (Data Availability). Conditional bonuses remain at zero in the base scenario: the same test, run on the 46 cards whose conditional bonus is worth at least 1 C, found no factor to adopt (Sections VI-A and VII-D; preregistered as H10), but the 16 cards charged at about one half are understated by the score, and the player-controlled subgroup ($n = 9$) is a hypothesis only. Tagging a trigger as conditional is a rules-text judgement (17 of 162 cards) made after reviewing the ranked list, and the 13 units whose trigger tiers presuppose EMPOWER are scored against their printed cost pending a separate decision. *Sample size.* Most parameters rest on 3–15 cards; the statistical resolution is about 0.3 C, and differences below that cannot be resolved. *Low-confidence prices* (Table XIV) are flagged explicitly. *Coverage.* Gear is priced with an expected-uses parameter whose condition factors (1, 0.5, 0.25) are judgemental and which the ablation does not test; 38 gear cards are not fully priced. Battlefields and legends have per-use yields but no expected-use model and are not ranked; 3 legends are deferred. Of the 110 units not fully priced, 51 carry an unpriced effect (mostly one-off mechanics or "can't be chosen") and 59 are deferred (all or indefinite targets, gear- or legend-dependent effects, might-dependent

TABLE XIV
LOW-CONFIDENCE PRICES AND THEIR TREATMENT

Item	Treatment	Reason
TEMPORARY	not priced	≈ -2.65 C from two Sprite cards only
Look at opponent's hand, discard any card	4 C	Mindsplitter: 4 C by cost-minus-might, 6.5 C by budget method
Move an enemy unit to its base	2 C	Isolate single anchor; Maddened Marauder (OGN-191) recovers 1 C
Prevent damage	2 C / 4 C	One anchor each
Tokens from Deathknell	token $\times 0.5$	3 samples, mean 0.70 (Machine Evangel 0.95, Ferrous Forerunner 0.44, Carrion Dredger 0.71), dispersed
Flow keyword at $p = 0$	0.2 C (0–0.6)	4 direct cards; per-card noise ± 1.4 C; bootstrap 90% interval of the adopted rule 0.00–0.28
Destroy a gear	2.75 C	Four anchors bimodal (Brittle Steel 4.0, Disarming Rake 3.5, Salvage 2.0, Turn to Dust 2.0); median bootstrap interval 2.0–4.0
Stun, unrestricted	4 C	Rune Prison single anchor; Back Off implies ≤ 3, Zenith Blade ≥ 6
Sacrifice an ally (additional cost)	-2 C	Cruel Patron single anchor; Sacrifice implies -5
Might budget at 7–8 C	$B(c)$ kept	Above 5 C the curve rests on Mega-Mech and Mountain Drake; no vanilla 8 C unit; four fully priced 8 C units total 6.5–8 C against $\lfloor B(8) \rfloor = 9$
Budget-curve slopes 1.1 / 1.33	kept	Identified only to $[1.00, 1.20)$ and $[1.25, 1.375)$ under the floor convention; $\lfloor B(c) \rfloor$ unchanged at 1–9 C; the unrounded slack (keyword-rounding boundary, $1.10\times$ multiplier) depends on the choice
Spell w/o timing keyword: $\pm X$ might this turn, stun	$\times 0.50$	Onslaught is the only usable sample; $-X$ might is included by rule, and its printed anchors (Frostcoat Cub, Blastcone Fae) point to full price

effects, play from the trash, zero-cost tokens, rule changes, free play, TEMPORARY or copies, and one card by decision). Of the 95 spells, 19 are partial and 76 deferred: 21 with all or indefinite targets, 16 might-dependent, 11 counters, 7 symmetric, 5 control theft, and 16 involving a ban, gear, an effect with no current price, play from the trash, free play, a rule change, "can't be chosen" or TEMPORARY (Table I). A possible discount for very expensive spells was examined but not adopted for lack of samples.

IX. CONCLUSION

We have shown that a CCG designer's internal costing model can be recovered from printed cards alone by treating pricing as an inverse hedonic problem: ZHIZHI inverts effect prices from zero-residual anchor pairs, imputes every card's value from its text without refitting, and supplements the linear core with three mechanism models drawn from behavioural economics and order statistics; the AMBER score reports the result per card. For Riftbound the recovered table reproduces printed costs for the central mass of 497 cards (median AMBER score 1.00, IQR 0.86–1.07; nominal IQR 0.75–1.00), predicts held-out cards, and exposes design rules—a 50% booking of repeatable contingent value against a zero booking of most one-off conditional bonuses, free play-timing, near-free on-play effects on cheap units, a 1 C same-target complementarity premium for spells that reposition and enhance one unit—that are not stated anywhere in the published rules; booking repeatable triggers at half an activation (a quarter under an independent condition) improves prediction card by card under leave-one-out re-estimation of the prices, with the factors themselves fixed (Section VII-D). An ablation over nested models shows that the cost-controlled on-play adjustments account for nearly all of the error reduction and retain it under leave-one-out cross-validation, whereas the hand-card and selection layers change few base values and, on the cards they change, can be separated from the layer below neither by squared error nor by the exact-cost hit rate at the game's 1 C granularity (Section VII-C, Table XI); a review of every price against its own anchors under leave-one-out and bootstrap resampling found none that could be moved by more than the model's resolution (Section VII-E). Future work will test the gear model's expected-use parameter under the same ablation, add an expected-use model for battlefields and legends, and produce the pool-level summary (mean value by domain, type and set, and per-keyword prices by domain) needed to test whether domains are systematically over- or under-charged for the same effect.

DATA AVAILABILITY

The card export, per-card valuations (base and full-trigger values, trigger bins, Flow regimes), per-card AMBER and nominal AMBER scores, the 162-card scope list, the 17-card conditional-trigger list and the 54-card conditional-bonus list with its per-card f^*, the anchor table, the budget-curve check and the audit change log will be released as a separate dataset on Zenodo; its DOI will be cited in the next version of this paper. A blind test of the frozen model on the fifth set, *Radiance*, is preregistered at `https://doi.org/10.5281/zenodo.22928382` (hypotheses H1–H8, registered 24 September 2026, use the nominal score). An addendum registering H9, the out-of-sample test of the trigger factors of Section III-D, and H10, the zero booking of one-off conditional bonuses (Section VI-A), was added to the same record as version 2 (`https://doi.org/10.5281/zenodo.22950713`, 25 September 2026) before any *Radiance* card was priced; the frozen model is unchanged by it.

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